

Geometry of p –harmonic maps between Riemannian manifolds

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Introduction

Let $C^\infty(M, N)$ denote the space of smooth maps $\varphi : (M, g) \longrightarrow (N, h)$ between Riemannian manifolds. For any compact domain $D \subset M$, the energy of φ over D is defined by

$$E(\varphi) = \frac{1}{2} \int_D |\mathrm{d}\varphi|^2 v_g, \quad (0.1)$$

where $|\mathrm{d}\varphi|$ is the Hilbert-Schmidt norm of the differential $\mathrm{d}\varphi$, and v_g is the volume element of (M, g) . A map φ is called harmonic if it is a critical point of this energy on every compact domain $D \subset M$. If $\{\varphi_t\}_{t \in (-\varepsilon, \varepsilon)}$ is a smooth variation of φ , then its first variation is

$$\left. \frac{\mathrm{d}}{\mathrm{d}t} E(\varphi_t) \right|_{t=0} = - \int_D \langle v, \tau(\varphi) \rangle v_g, \quad (0.2)$$

where

$$v = \left. \frac{\partial \varphi_t}{\partial t} \right|_{t=0}$$

denotes the corresponding variation vector field. The section $\tau(\varphi)$ is the tension field of φ and is given by

$$\tau(\varphi) = \operatorname{trace}_g \nabla \mathrm{d}\varphi.$$

A natural higher-order analogue of harmonicity is obtained by considering the square of the norm of the tension field. The bienergy of φ is

$$E_2(\varphi) = \frac{1}{2} \int_D |\tau(\varphi)|^2 v_g. \quad (0.3)$$

A map is called biharmonic when it is a critical point of the bienergy. Equivalently, it satisfies the Euler-Lagrange equation associated with (0.3), namely

$$\tau_2(\varphi) = -\Delta^\varphi \tau(\varphi) - \operatorname{trace}_g R^N(\mathrm{d}\varphi, \tau(\varphi)) \mathrm{d}\varphi = 0. \quad (0.4)$$

This class of maps has been studied, among others, by J. Eells, J. H. Sampson, L. Lemaire, A. Lichnerowicz, and G. Y. Jiang; see, for instance, [10, 9, 17, 14].

A submanifold of a Riemannian manifold is said to be harmonic, or minimal, when the isometric immersion defining it is a harmonic map. It is called biharmonic when the corresponding isometric immersion is a biharmonic map. A biharmonic submanifold that is not minimal will be called a proper biharmonic submanifold (see [3]).

The variational problem obtained by fixing a map and varying the Riemannian metric on a compact domain leads naturally to the stress-energy tensor. This point of view is useful not only for harmonic maps but also for their higher-order and p -energy analogues.

For $p \geq 2$, a p -harmonic map is a critical point of the p -energy functional

$$E_p(\varphi) = \frac{1}{p} \int_D |\mathrm{d}\varphi|^p v_g. \quad (0.5)$$

If $\{\varphi_t\}_{t \in (-\varepsilon, \varepsilon)}$ is a smooth variation of φ supported in D , then

$$\left. \frac{d}{dt} E_p(\varphi_t) \right|_{t=0} = - \int_D \langle \tau_p(\varphi), v \rangle v_g, \quad (0.6)$$

where $v = \partial\varphi_t/\partial t|_{t=0}$ is the variation vector field and

$$\tau_p(\varphi) = \operatorname{div}^M(|d\varphi|^{p-2} d\varphi). \quad (0.7)$$

The ordinary tension field can be written locally as

$$\tau(\varphi) = \operatorname{trace}_g \nabla d\varphi = \sum_{i=1}^m \left\{ \nabla_{e_i}^\varphi d\varphi(e_i) - d\varphi(\nabla_{e_i}^M e_i) \right\}, \quad (0.8)$$

where $\{e_1, \dots, e_m\}$ is an orthonormal frame on (M, g) , $m = \dim M$, ∇^M is the Levi-Civita connection of (M, g) , and ∇^φ denotes the pull-back connection on $\varphi^{-1}TN$ (see [2]). If $|d\varphi|_x \neq 0$ for every $x \in M$, then φ is p -harmonic if and only if

$$|d\varphi|^{p-2} \tau(\varphi) + (p-2)|d\varphi|^{p-3} d\varphi(\operatorname{grad}^M |d\varphi|) = 0, \quad (0.9)$$

see [1, 4, 11].

The p -biharmonic theory is obtained by applying the same idea to the p -tension field. More precisely, the p -bienergy functional is

$$E_{2,p}(\varphi) = \frac{1}{2} \int_D |\tau_p(\varphi)|^2 v_g. \quad (0.10)$$

A map φ is called p -biharmonic if it is a critical point of (0.10). Equivalently, it satisfies the Euler-Lagrange equation (see [20])

$$\begin{aligned} \tau_{2,p}(\varphi) = & -|d\varphi|^{p-2} \operatorname{trace}_g R^N(\tau_p(\varphi), d\varphi) d\varphi \\ & - \operatorname{trace}_g \nabla^\varphi(|d\varphi|^{p-2} \nabla^\varphi \tau_p(\varphi)) \\ & - (p-2) \operatorname{trace}_g \nabla \langle \nabla^\varphi \tau_p(\varphi), d\varphi \rangle |d\varphi|^{p-4} d\varphi = 0. \end{aligned} \quad (0.11)$$

For an orthonormal frame $\{e_1, \dots, e_m\}$ on (M, g) , the terms in (0.11) are understood as follows:

$$\begin{aligned} \operatorname{trace}_g R^N(\tau_p(\varphi), d\varphi) d\varphi &= \sum_{i=1}^m R^N(\tau_p(\varphi), d\varphi(e_i)) d\varphi(e_i), \\ \operatorname{trace}_g \nabla^\varphi(|d\varphi|^{p-2} \nabla^\varphi \tau_p(\varphi)) &= \sum_{i=1}^m \left(\nabla_{e_i}^\varphi(|d\varphi|^{p-2} \nabla_{e_i}^\varphi \tau_p(\varphi)) - |d\varphi|^{p-2} \nabla_{\nabla_{e_i}^M e_i}^\varphi \tau_p(\varphi) \right), \\ \langle \nabla^\varphi \tau_p(\varphi), d\varphi \rangle &= \sum_{i=1}^m \langle \nabla_{e_i}^\varphi \tau_p(\varphi), d\varphi(e_i) \rangle, \end{aligned}$$

and

$$\begin{aligned} & \operatorname{trace}_g \nabla \langle \nabla^\varphi \tau_p(\varphi), d\varphi \rangle |d\varphi|^{p-4} d\varphi \\ &= \sum_{i=1}^m \left(\nabla_{e_i}^\varphi \langle \nabla^\varphi \tau_p(\varphi), d\varphi \rangle |d\varphi|^{p-4} d\varphi(e_i) - \langle \nabla^\varphi \tau_p(\varphi), d\varphi \rangle |d\varphi|^{p-4} d\varphi(\nabla_{e_i}^M e_i) \right). \end{aligned}$$

When $p = 2$, the p -energy and p -bienergy reduce to the classical energy and bienergy, and their critical points are the usual harmonic and biharmonic maps. Moreover, every p -harmonic map is automatically p -biharmonic. In particular, if (M, g) is a compact orientable Riemannian manifold

without boundary and (N, h) has non-positive sectional curvature, then every p -biharmonic map from (M, g) into (N, h) is p -harmonic.

A submanifold of a Riemannian manifold is called p -biharmonic if its defining isometric immersion is a p -biharmonic map. It is called proper p -biharmonic if it is p -biharmonic but not p -harmonic.

This thesis is mainly concerned with p -biharmonic submanifolds in space forms, as well as p -biharmonic hypersurfaces in Einstein spaces and conformally flat spaces. The first chapter fixes notation and recalls basic facts on Riemannian manifolds that will be used throughout the work. The second chapter introduces Riemannian submanifolds. The third chapter reviews the theory of harmonic and biharmonic maps. In the fourth chapter, the definitions of p -harmonic and p -biharmonic maps between Riemannian manifolds are developed further, together with several properties of the stress p -bienergy tensor (see [23]). The fifth chapter presents methods for constructing proper p -biharmonic hypersurfaces in Einstein spaces and conformally flat spaces, along with fundamental results and examples of proper p -biharmonic submanifolds.

Chapter 1

Elementary Differential Geometry

This chapter fixes the basic notation from differential and Riemannian geometry used throughout the thesis. We recall differentiable manifolds and submanifolds, tangent and cotangent bundles, Riemannian metrics, linear connections, curvature, and the standard differential operators on a Riemannian manifold.

1.1 Differentiable Manifolds and Submanifolds

The material in this section follows standard references such as [5], [16], [26], [27], and [28].

1.1.1 The definition of a differentiable manifold

Definition 1.1.1 (Chart). A topological space M is said to be *locally Euclidean of dimension n* if every point $p \in M$ has a neighbourhood U for which there is a homeomorphism

$$\varphi : U \longrightarrow \varphi(U) \subset \mathbb{R}^n,$$

where $\varphi(U)$ is open in $\mathbb{R}^n$. The pair (U, φ) is called a *chart*.

Definition 1.1.2 (Topological manifold). A *topological manifold of dimension n* is a Hausdorff space which is locally Euclidean of dimension n and has a countable basis of open sets.

Definition 1.1.3 (Compatible charts). Two charts (U, φ) and (V, ψ) on a topological space are said to be C^∞ -compatible if, whenever $U \cap V \neq \emptyset$, the transition maps

$$\varphi \circ \psi^{-1} : \psi(U \cap V) \longrightarrow \varphi(U \cap V), \quad \psi \circ \varphi^{-1} : \varphi(U \cap V) \longrightarrow \psi(U \cap V)$$

are smooth.

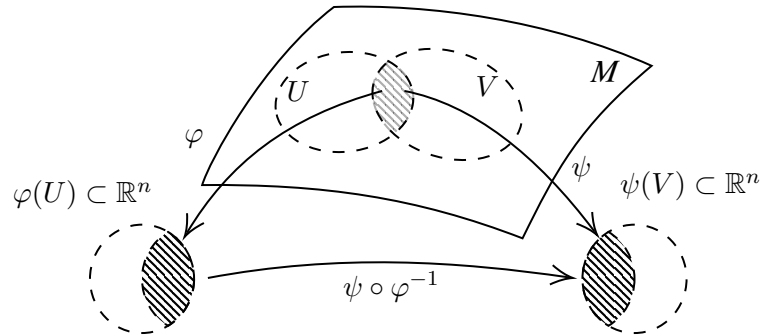


Figure 1.1: Transition maps are defined on the coordinate images of chart overlaps.

Definition 1.1.4 (Atlas). A C^∞ atlas on a locally Euclidean space M is a collection

$$\mathcal{A} = \{(U_\alpha, \varphi_\alpha)\}_\alpha$$

of pairwise C^∞ -compatible charts such that $M = \bigcup_\alpha U_\alpha$.

Definition 1.1.5 (Smooth manifold). A C^∞ -manifold, or differentiable manifold, is a topological manifold equipped with a C^∞ atlas.

Theorem 1.1.6. Let M and N be C^∞ -manifolds of dimensions m and n . Then $M \times N$ is a C^∞ -manifold of dimension $m+n$. If $(U_\alpha, \varphi_\alpha)$ and (V_β, ψ_β) are charts on M and N , respectively, then $(U_\alpha \times V_\beta, \varphi_\alpha \times \psi_\beta)$ are product charts, where $(\varphi_\alpha \times \psi_\beta)(p, q) = (\varphi_\alpha(p), \psi_\beta(q)) \in \mathbb{R}^{m+n} = \mathbb{R}^m \times \mathbb{R}^n$.

1.1.2 Differentiable functions and mappings

Let f be a real-valued function defined on an open subset W_f of an n -dimensional smooth manifold M . If (U, φ) is a chart and $W_f \cap U \neq \emptyset$, the local expression of f is given by

$$\tilde{f} = f \circ \varphi^{-1} : \varphi(W_f \cap U) \longrightarrow \mathbb{R}.$$

Definition 1.1.7 (Smooth function). A function $f : W_f \rightarrow \mathbb{R}$ is of class C^∞ if, for every $p \in W_f$, there is a chart (U, φ) around p such that $f \circ \varphi^{-1}$ is smooth on $\varphi(W_f \cap U)$.

Definition 1.1.8 (Smooth mapping). Let M and N be smooth manifolds. A map $F : M \rightarrow N$ is smooth if, for every $p \in M$, there are coordinate neighbourhoods (U, φ) of p and (V, ψ) of $F(p)$, with $F(U) \subset V$, such that

$$\psi \circ F \circ \varphi^{-1} : \varphi(U) \longrightarrow \psi(V)$$

is a smooth map between open subsets of Euclidean spaces.

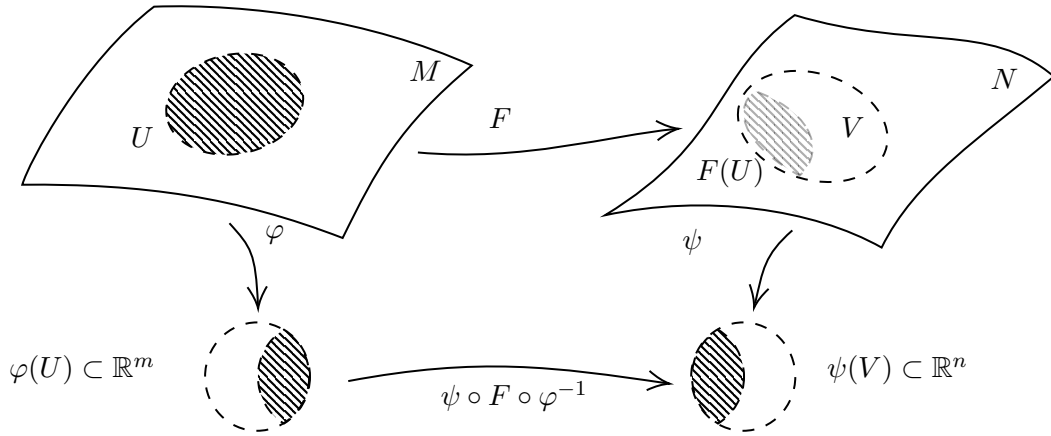


Figure 1.2: A smooth mapping is checked through its coordinate representation.

Definition 1.1.9 (Diffeomorphism). A smooth map $F : M \rightarrow N$ between smooth manifolds is a *diffeomorphism* if it is a homeomorphism and F^{-1} is also smooth. In this case M and N are called diffeomorphic.

1.1.3 Rank, immersions, and submersions

Let $F : M^m \rightarrow N^n$ be a smooth map and let $p \in M$. Choose charts (U, φ) around p and (V, ψ) around $F(p)$ such that $F(U) \subset V$. The coordinate representative of F is

$$\tilde{F} = \psi \circ F \circ \varphi^{-1} : \varphi(U) \longrightarrow \psi(V).$$

Definition 1.1.10 (Rank). The *rank* of F at p is the rank of the Euclidean map $\widetilde{F}$ at the point $\varphi(p)$. If

$$\widetilde{F}(x^1, \dots, x^m) = (f^1(x), \dots, f^n(x)),$$

then $\text{rank}_p F$ is the rank of the Jacobian matrix

$$\left(\frac{\partial f^\alpha}{\partial x^i} \right)_{\substack{\alpha=1, \dots, m \\ i=1, \dots, n}}$$

evaluated at $\varphi(p)$.

Definition 1.1.11 (Immersion and submersion). A map $F : M^m \rightarrow N^n$ is an *immersion* if $\text{rank } F = m$ everywhere. It is a *submersion* if $\text{rank } F = n$ everywhere. Thus an immersion can exist only if $m \leq n$, while a submersion can exist only if $n \leq m$.

If F is an injective immersion, its image may be endowed with a topology and smooth structure so that F becomes a diffeomorphism onto this image. This image is called an immersed submanifold. In general, the topology and smooth structure of an immersed submanifold depend on the map F and need not be the subspace topology inherited from N .

Definition 1.1.12 (Embedding). An *embedding* is a one-to-one immersion $F : M \rightarrow N$ that is a homeomorphism from M onto its image $F(M)$, where $F(M)$ carries the subspace topology from N . The image of an embedding is called an embedded submanifold.

Theorem 1.1.13. *If $F : M \rightarrow N$ is an immersion, then every point $p \in M$ has a neighbourhood U such that $F|_U$ is an embedding of U into N .*

1.1.4 Submanifolds

Definition 1.1.14 (Submanifold property). Let N be a smooth n -manifold. A subset $M \subset N$ has the *m -submanifold property* if every point $p \in M$ has a coordinate neighbourhood (U, φ) in N , with coordinates $(x^1, \dots, x^n)$, such that

- (i) $\varphi(p) = (0, \dots, 0)$;
- (ii) $\varphi(U) = C_\varepsilon^n(0) = \{x \in \mathbb{R}^n : |x^i| < \varepsilon, i = 1, \dots, n\}$;
- (iii) $\varphi(U \cap M) = \{x \in C_\varepsilon^n(0) : x^{m+1} = \dots = x^n = 0\}$.

Such coordinates are called preferred coordinates relative to M .

Lemma 1.1.15. *If $M \subset N$ has the m -submanifold property, then M , with the relative topology, is a topological m -manifold. Moreover, the preferred coordinate systems induce smooth local coordinates on M , compatible on overlaps, and give a smooth structure for which the inclusion $i : M \hookrightarrow N$ is an embedding.*

Definition 1.1.16 (Regular submanifold). A *regular submanifold* of a smooth manifold N is a subset $M \subset N$ with the submanifold property, equipped with the smooth structure determined by its preferred coordinates.

Theorem 1.1.17. *Let $F : M' \rightarrow N$ be an embedding of a smooth m -manifold M' into a smooth n -manifold N . Then $M = F(M')$ has the m -submanifold property and hence is a regular submanifold. As a regular submanifold, it is diffeomorphic to M' via $F : M' \rightarrow M$.*

Theorem 1.1.18. *If $F : M \rightarrow N$ is a one-to-one immersion and M is compact, then F is an embedding. Consequently, $F(M)$ is a regular submanifold of N .*

Theorem 1.1.19. *Let M and N be smooth manifolds of dimensions m and n , and let $F : M \rightarrow N$ be a smooth map of constant rank k . If $q \in F(M)$, then $F^{-1}(q)$ is a closed regular submanifold of M of dimension $m - k$.*

Corollary 1.1.20. *Let $F : M \rightarrow N$ be a smooth map with $\dim N = n \leq m = \dim M$. If $\text{rank } F = n$ at every point of $A = F^{-1}(q)$, then A is a closed regular submanifold of M .*

Example 1.1.21 (Sphere). Consider

$$F : \mathbb{R}^n \longrightarrow \mathbb{R}, \quad F(x) = \sum_{i=1}^n (x^i)^2.$$

The map F has rank 1 on $\mathbb{R}^n \setminus \{0\}$, and $F^{-1}(1) = S^{n-1}$. Therefore S^{n-1} is a regular submanifold of dimension $n - 1$.

Example 1.1.22 (Torus). For $a > b > 0$, define

$$F : \mathbb{R}^3 \longrightarrow \mathbb{R}, \quad F(x, y, z) = \left(a - \sqrt{x^2 + y^2}\right)^2 + z^2.$$

The level set $F^{-1}(b^2)$ has rank 1 at each of its points. Hence $F^{-1}(b^2)$ is a two-dimensional submanifold of $\mathbb{R}^3$, namely the standard torus.

1.2 Vector Fields on Smooth Manifolds

1.2.1 The tangent space at a point

Let M be a smooth manifold of dimension n . A germ of a smooth function at a point $p \in M$ is an equivalence class of smooth functions defined near p , where two functions are identified if they agree on some neighbourhood of p . We denote by $C_p^\infty(M)$ the algebra of germs of smooth real-valued functions at p .

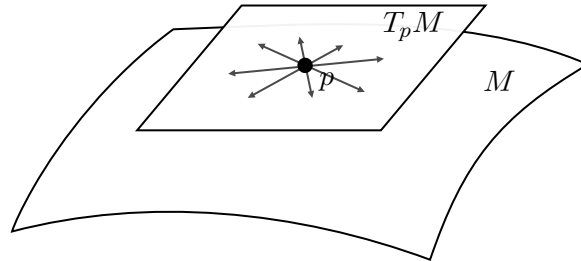
Definition 1.2.1 (Tangent vector). The tangent space $T_p M$ is the set of all maps

$$X_p : C_p^\infty(M) \longrightarrow \mathbb{R}$$

which satisfy, for all $\alpha, \beta \in \mathbb{R}$ and all $f, g \in C_p^\infty(M)$,

- (i) $X_p(\alpha f + \beta g) = \alpha X_p(f) + \beta X_p(g)$;
- (ii) $X_p(fg) = X_p(f)g(p) + f(p)X_p(g)$.

An element of $T_p M$ is called a tangent vector to M at p .



Lemma 1.2.2. *Let $X_p \in T_p M$.*

- (1) *If f and g agree on a neighbourhood of p , then $X_p(f) = X_p(g)$.*
- (2) *If h is constant on a neighbourhood of p , then $X_p(h) = 0$.*

Definition 1.2.3 (Coordinate vectors). Let $(U, \varphi) = (U, x^1, \dots, x^n)$ be a coordinate system around p . For $f \in C^\infty(M)$, define

$$\left. \frac{\partial}{\partial x^i} \right|_p (f) = \frac{\partial(f \circ \varphi^{-1})}{\partial u^i}(\varphi(p)), \quad i = 1, \dots, n,$$

where $(u^1, \dots, u^n)$ are the standard coordinates on $\mathbb{R}^n$.

Theorem 1.2.4. For a coordinate system $(x^1, \dots, x^n)$ around p , the coordinate vectors

$$\left. \frac{\partial}{\partial x^1} \right|_p, \dots, \left. \frac{\partial}{\partial x^n} \right|_p$$

form a basis of $T_p M$. Hence each $X_p \in T_p M$ can be written uniquely as

$$X_p = \sum_{i=1}^n X_p(x^i) \left. \frac{\partial}{\partial x^i} \right|_p.$$

Definition 1.2.5 (Differential of a map). Let $\varphi : M \rightarrow N$ be a smooth map. The differential of φ at p is the linear map $d\varphi_p : T_p M \rightarrow T_{\varphi(p)} N$ defined by

$$d\varphi_p(X_p)(g) = X_p(g \circ \varphi), \quad g \in C_{\varphi(p)}^\infty(N).$$

1.2.2 Vector fields

Definition 1.2.6 (Tangent bundle). The tangent bundle of a smooth manifold M is the disjoint union of its tangent spaces:

$$TM = \bigcup_{p \in M} T_p M.$$

It carries a natural smooth manifold structure, with $\dim TM = 2 \dim M$.

Definition 1.2.7 (Vector bundle). Let E and M be smooth manifolds, and let $\pi : E \rightarrow M$ be a smooth surjective map. The fibre over $p \in M$ is $E_p = \pi^{-1}(p)$. In particular, the natural projection $\pi : TM \rightarrow M$, given by $\pi(p, v) = p$, makes TM into a smooth vector bundle over M .

Definition 1.2.8 (Vector field). A section of a vector bundle $\pi : E \rightarrow M$ is a map $s : M \rightarrow E$ such that $\pi \circ s = \text{Id}_M$. A vector field on M is a section of the tangent bundle, that is, an assignment

$$p \mapsto X_p \in T_p M.$$

The vector field is smooth if it is smooth as a map $M \rightarrow TM$. The set of smooth vector fields on M is denoted by $\Gamma(TM)$.

1.3 Riemannian Manifolds

1.3.1 Tangent covectors

Definition 1.3.1 (Covector). For $p \in M$, the cotangent space $T_p^* M$ is the dual vector space of $T_p M$. Thus a covector $\sigma_p \in T_p^* M$ is a linear map

$$\sigma_p : T_p M \rightarrow \mathbb{R}.$$

If $e_{1p}, \dots, e_{np}$ is a basis of $T_p M$, its dual basis $\omega_p^1, \dots, \omega_p^n$ is determined by

$$\omega_p^i(e_{jp}) = \delta_j^i.$$

Every covector can be written as

$$\sigma_p = \sum_{i=1}^n \sigma_p(e_{ip}) \omega_p^i.$$

Definition 1.3.2 (Covector field). A smooth covector field σ on M assigns to every $p \in M$ a covector $\sigma_p \in T_p^*M$ in such a way that, in any coordinate frame $e_1, \dots, e_n$ on a chart U , the functions $\sigma(e_i)$ are smooth on U .

Remark 1.3.3. The cotangent bundle is denoted by

$$T^*M = \bigcup_{p \in M} T_p^*M.$$

If $(U, x^1, \dots, x^n)$ is a local chart, then $dx^1, \dots, dx^n$ form the local basis of T^*M dual to $\frac{\partial}{\partial x^1}, \dots, \frac{\partial}{\partial x^n}$.

Definition 1.3.4 (Tensor field). For $p \in M$, define

$$T_p^{(r,s)}M = \underbrace{T_pM \otimes \dots \otimes T_pM}_{r \text{ times}} \otimes \underbrace{T_p^*M \otimes \dots \otimes T_p^*M}_{s \text{ times}}.$$

The tensor bundle is $T^{(r,s)}M = \bigcup_{p \in M} T_p^{(r,s)}M$. A tensor field of type (r, s) is a smooth section of this bundle.

Example 1.3.5. A function is a $(0, 0)$ -tensor, a vector field is a $(1, 0)$ -tensor, and a differential 1-form is a $(0, 1)$ -tensor.

1.3.2 Riemannian metrics

Definition 1.3.6 (Riemannian metric). A Riemannian metric on a smooth manifold M is a smooth $(0, 2)$ -tensor field g which is bilinear, symmetric, and positive definite. Thus at each $p \in M$,

$$g_p : T_pM \times T_pM \longrightarrow \mathbb{R}$$

is an inner product. The pair (M, g) is called a Riemannian manifold.

1.3.3 Orientability

Definition 1.3.7 (Orientation of a vector space). An orientation on a finite-dimensional real vector space is a choice of one of the two equivalence classes of ordered bases, where two bases are equivalent if the change-of-basis matrix has positive determinant. Bases in the chosen class are called positively oriented.

Lemma 1.3.8. Let $\Omega \neq 0$ be an alternating covariant tensor of order $n = \dim V$ on a vector space V . If $e_1, \dots, e_n$ is a basis and

$$v_i = \sum_{j=1}^n \alpha_i^j e_j,$$

then

$$\Omega(v_1, \dots, v_n) = \det(\alpha_i^j) \Omega(e_1, \dots, e_n).$$

Corollary 1.3.9. A non-vanishing element $\Omega \in \Lambda^n(V^*)$ determines an orientation of V . Two such forms determine the same orientation if and only if they differ by multiplication by a positive function.

Definition 1.3.10 (Orientable manifold). A smooth n -manifold M is orientable if it admits a nowhere vanishing smooth n -form Ω . Such a form determines an orientation of M .

Theorem 1.3.11. A manifold M is orientable if and only if it admits an atlas whose coordinate changes are coherently oriented, that is, have positive Jacobian determinant on overlaps.

Theorem 1.3.12. *Let (M, g) be an orientable Riemannian manifold. For each choice of orientation, there is a unique n -form Ω which determines that orientation and has value $+1$ on every positively oriented orthonormal frame.*

Example 1.3.13 (The sphere). The sphere

$$S^n = \{x \in \mathbb{R}^{n+1} : \|x\| = 1\}$$

is orientable. Using the stereographic projections from the north pole $N = (0, \dots, 0, 1)$ and the south pole $S = (0, \dots, 0, -1)$,

$$\varphi_N(u) = \left(\frac{u^1}{1 - u^{n+1}}, \dots, \frac{u^n}{1 - u^{n+1}} \right),$$

and

$$\varphi_S(u) = \left(\frac{u^1}{1 + u^{n+1}}, \dots, \frac{u^n}{1 + u^{n+1}} \right),$$

one obtains a smooth atlas on S^n . The inverse charts are

$$\varphi_N^{-1}(x) = \left(\frac{2x^1}{\|x\|^2 + 1}, \dots, \frac{2x^n}{\|x\|^2 + 1}, \frac{\|x\|^2 - 1}{\|x\|^2 + 1} \right),$$

$$\varphi_S^{-1}(y) = \left(\frac{2y^1}{\|y\|^2 + 1}, \dots, \frac{2y^n}{\|y\|^2 + 1}, -\frac{\|y\|^2 - 1}{\|y\|^2 + 1} \right).$$

The transition maps are

$$\varphi_S \circ \varphi_N^{-1}(x) = \frac{x}{\|x\|^2}, \quad \varphi_N \circ \varphi_S^{-1}(y) = \frac{y}{\|y\|^2},$$

for $x, y \in \mathbb{R}^n \setminus \{0\}$.

1.3.4 Integration on manifolds

On an oriented n -manifold one integrates compactly supported n -forms. Let $\Omega_c^k(M)$ denote the vector space of smooth compactly supported k -forms on M . If (U, φ) is an oriented chart and $\omega \in \Omega_c^n(U)$, define

$$\int_U \omega := \int_{\varphi(U)} (\varphi^{-1})^* \omega. \quad (1.1)$$

This definition is independent of the oriented chart. Indeed, if (U, ψ) is another oriented chart, then the change of variables formula gives

$$\int_{\varphi(U)} (\varphi^{-1})^* \omega = \int_{\psi(U)} (\psi^{-1})^* \omega.$$

For a compactly supported n -form ω on M , choose a partition of unity $\{\rho_\alpha\}$ subordinate to an oriented atlas $\{U_\alpha\}$. Since ω has compact support, only finitely many $\rho_\alpha \omega$ are non-zero, and we define

$$\int_M \omega := \sum_\alpha \int_{U_\alpha} \rho_\alpha \omega.$$

1.3.5 Manifolds with boundary

Definition 1.3.14 (Smooth manifold with boundary). A smooth manifold with boundary is a Hausdorff space M with a countable basis of open sets and a differentiable structure whose charts take values in the half-space

$$H^n = \{(x^1, \dots, x^n) \in \mathbb{R}^n : x^1 \geq 0\}.$$

The transition maps are diffeomorphisms between open subsets of H^n , and the atlas is maximal with respect to this property.

Remark 1.3.15. The boundary of the half-space is

$$\partial H^n = \{(x^1, \dots, x^n) \in \mathbb{R}^n : x^1 = 0\}.$$

A point $p \in M$ belongs to the boundary ∂M if, in one (and hence in every) boundary chart, its image lies in ∂H^n . The complement $M \setminus \partial M$ is the interior of M . If $\partial M = \emptyset$, then M is a manifold without boundary.

1.4 Connections on Riemannian Manifolds

1.4.1 Linear connections

Definition 1.4.1 (Linear connection). A linear connection on a smooth manifold M is a map $\nabla : \Gamma(TM) \times \Gamma(TM) \longrightarrow \Gamma(TM)$ given by $(X, Y) \longmapsto \nabla_X Y$, such that, for all $X, Y, Z \in \Gamma(TM)$ and $f \in C^\infty(M)$,

- (1) $\nabla_X(Y + Z) = \nabla_X Y + \nabla_X Z$;
- (2) $\nabla_X(fY) = f\nabla_X Y + X(f)Y$;
- (3) $\nabla_{X+fY}Z = \nabla_X Z + f\nabla_Y Z$.

The vector field $\nabla_X Y$ is the covariant derivative of Y in the direction of X .

Definition 1.4.2 (Parallel vector field). A vector field $Y \in \Gamma(TM)$ is parallel with respect to ∇ if

$$\nabla_X Y = 0 \quad \text{for every } X \in \Gamma(TM).$$

Definition 1.4.3 (Metric connection). A connection ∇ is compatible with a Riemannian metric g if $\nabla g = 0$, that is,

$$X(g(Y, Z)) = g(\nabla_X Y, Z) + g(Y, \nabla_X Z)$$

for all $X, Y, Z \in \Gamma(TM)$.

1.4.2 Torsion tensor

Definition 1.4.4 (Torsion). The torsion tensor of a connection ∇ on TM is

$$T(X, Y) = \nabla_X Y - \nabla_Y X - [X, Y].$$

The connection is torsion-free if $T = 0$, equivalently

$$[X, Y] = \nabla_X Y - \nabla_Y X.$$

Remark 1.4.5. The torsion tensor is antisymmetric: $T(X, Y) = -T(Y, X)$.

1.4.3 Levi-Civita connection

Definition 1.4.6 (Levi-Civita connection). Let (M, g) be a Riemannian manifold. The Levi-Civita connection is the connection determined by the Koszul formula

$$2g(\nabla_X Y, Z) = X(g(Y, Z)) + Y(g(Z, X)) - Z(g(X, Y)) \quad (1.2)$$

$$+ g(Z, [X, Y]) + g(Y, [Z, X]) - g(X, [Y, Z]). \quad (1.3)$$

Theorem 1.4.7. *On a Riemannian manifold (M, g) , there exists a unique linear connection which is compatible with g and torsion-free. It is the Levi-Civita connection.*

Proposition 1.4.8 (Christoffel symbols). *Let $(U, x^1, \dots, x^m)$ be a coordinate chart on a Riemannian manifold (M, g) and write $\partial_i = \partial/\partial x^i$. The Christoffel symbols are defined by*

$$\nabla_{\partial_i} \partial_j = \sum_{k=1}^m \Gamma_{ij}^k \partial_k.$$

They are given by

$$\Gamma_{ij}^k = \frac{1}{2} \sum_{\ell=1}^m g^{k\ell} \left(\frac{\partial g_{j\ell}}{\partial x^i} + \frac{\partial g_{i\ell}}{\partial x^j} - \frac{\partial g_{ij}}{\partial x^\ell} \right),$$

where $g_{ij} = g(\partial_i, \partial_j)$ and $(g^{ij}) = (g_{ij})^{-1}$.

1.5 Induced Connection on the Pull-back Bundle

Definition 1.5.1 (Pull-back connection). Let $\varphi : M \rightarrow N$ be a smooth map between differentiable manifolds, and let ∇^N be a linear connection on N . The induced connection on the pull-back bundle $\varphi^{-1}TN$ is $\nabla^\varphi : \Gamma(TM) \times \Gamma(\varphi^{-1}TN) \rightarrow \Gamma(\varphi^{-1}TN)$, defined by $(X, V) \mapsto \nabla_X^\varphi V$, where locally one sets

$$\nabla_X^\varphi V = \nabla_{d\varphi(X)}^N \widetilde{V},$$

for any vector field $\widetilde{V}$ on N whose restriction along φ is V .

If $V = V^\alpha (\partial/\partial y^\alpha \circ \varphi)$ and $X = X^i \partial_i$, then

$$\nabla_X^\varphi V = X^i \left\{ \frac{\partial V^\gamma}{\partial x^i} + V^\alpha \frac{\partial \varphi^\beta}{\partial x^i} (\Gamma_{\alpha\beta}^\gamma \circ \varphi) \right\} \left(\frac{\partial}{\partial y^\gamma} \circ \varphi \right).$$

This local expression shows that the definition is independent of the auxiliary extension $\widetilde{V}$.

Definition 1.5.2 (φ -related vector fields). Let $\varphi : M \rightarrow N$ be smooth. Vector fields $X \in \Gamma(TM)$ and $\widetilde{X} \in \Gamma(TN)$ are φ -related if

$$d\varphi_p(X_p) = \widetilde{X}_{\varphi(p)} \quad \text{for all } p \in M.$$

In that case one writes $\widetilde{X} = d\varphi(X)$ along φ .

Proposition 1.5.3. *Let $\varphi : M \rightarrow N$ be smooth and let ∇^N be compatible with a Riemannian metric h on N . Then ∇^φ is compatible with the induced metric on $\varphi^{-1}TN$:*

$$X(h(V, W)) = h(\nabla_X^\varphi V, W) + h(V, \nabla_X^\varphi W).$$

Proof. Choose local vector fields $\widetilde{X}, \widetilde{V}, \widetilde{W}$ on N related to X, V, W along φ . Since ∇^N is metric,

$$\begin{aligned} X(h(V, W)) &= d\varphi(X)(h(\widetilde{V}, \widetilde{W})) \\ &= h(\nabla_{d\varphi(X)}^N \widetilde{V}, \widetilde{W}) + h(\widetilde{V}, \nabla_{d\varphi(X)}^N \widetilde{W}) \\ &= h(\nabla_X^\varphi V, W) + h(V, \nabla_X^\varphi W). \end{aligned}$$

□

Proposition 1.5.4. *If ∇^N is torsion-free, then for all $X, Y \in \Gamma(TM)$,*

$$\nabla_X^\varphi d\varphi(Y) = \nabla_Y^\varphi d\varphi(X) + d\varphi([X, Y]).$$

Proof. Let $V, W \in \Gamma(TN)$ be locally φ -related to X and Y . Since ∇^N is torsion-free,

$$\nabla_V^N W = \nabla_W^N V + [V, W].$$

Restricting along φ and using $[V, W] \circ \varphi = d\varphi([X, Y])$ gives the formula. □

1.6 Second Fundamental Form

Definition 1.6.1 (Second fundamental form of a map). Let $\varphi : (M, g) \rightarrow (N, h)$ be a smooth map between Riemannian manifolds. The second fundamental form of φ is the covariant derivative of the vector-valued 1-form $d\varphi$

$$\nabla d\varphi(X, Y) = \nabla_X^\varphi d\varphi(Y) - d\varphi(\nabla_X^M Y).$$

Definition 1.6.2 (Totally geodesic map). A map $\varphi : (M, g) \rightarrow (N, h)$ is totally geodesic if its second fundamental form vanishes identically.

Property 1.6.3. For a smooth map $\varphi : (M, g) \rightarrow (N, h)$, the second fundamental form is $C^\infty(M)$ -bilinear and symmetric:

$$\nabla d\varphi(X, Y) = \nabla d\varphi(Y, X).$$

Proposition 1.6.4 (Composition formula). *Let $\varphi : M \rightarrow N$ and $\psi : N \rightarrow P$ be smooth maps. Then*

$$\nabla d(\psi \circ \varphi) = d\psi(\nabla d\varphi) + \nabla d\psi(d\varphi, d\varphi).$$

Proof. For $X, Y \in \Gamma(TM)$,

$$\begin{aligned} \nabla d(\psi \circ \varphi)(X, Y) &= \nabla_X^{\psi \circ \varphi} d(\psi \circ \varphi)(Y) - d(\psi \circ \varphi)(\nabla_X^M Y) \\ &= \nabla_{d\varphi(X)}^\psi d\psi(d\varphi(Y)) - d\psi(d\varphi(\nabla_X^M Y)) \\ &= \nabla d\psi(d\varphi(X), d\varphi(Y)) + d\psi(\nabla d\varphi(X, Y)). \end{aligned}$$

□

Definition 1.6.5 (Geodesic frame). Let (M, g) be an m -dimensional Riemannian manifold. A local frame $\{e_i\}_{i=1}^m$ is called geodesic at $x \in M$ if it is orthonormal near x and

$$(\nabla_{e_i} e_j)_x = 0, \quad i, j = 1, \dots, m.$$

1.7 Curvature

1.7.1 Curvature tensor

Definition 1.7.1 (Curvature tensor). The curvature tensor of a connection is

$$R(X, Y)Z = [\nabla_X, \nabla_Y]Z - \nabla_{[X, Y]}Z = \nabla_X \nabla_Y Z - \nabla_Y \nabla_X Z - \nabla_{[X, Y]}Z.$$

The corresponding $(0, 4)$ -tensor is

$$R(X, Y, Z, W) = g(R(X, Y)Z, W).$$

Proposition 1.7.2. *On a Riemannian manifold, the curvature tensor satisfies*

- (1) $R(X, Y)Z = -R(Y, X)Z$;
- (2) $g(R(X, Y)Z, W) = g(R(Z, W)X, Y)$;
- (3) $g(R(X, Y)Z, Z) = 0$;
- (4) $R(X, Y)Z + R(Y, Z)X + R(Z, X)Y = 0$;
- (5) $(\nabla_X R)(Y, Z) + (\nabla_Y R)(Z, X) + (\nabla_Z R)(X, Y) = 0$.

1.7.2 Sectional curvature

Definition 1.7.3 (Sectional curvature). For $p \in M$ and linearly independent vectors $X, Y \in T_p M$, the sectional curvature is

$$K_p(X, Y) = \frac{g(R(X, Y)Y, X)}{g(X, X)g(Y, Y) - g(X, Y)^2}.$$

A Riemannian manifold has constant sectional curvature k if $K_p(X, Y) = k$ for every p and every two-plane $\text{span}\{X, Y\} \subset T_p M$.

Definition 1.7.4. Define the tensor R_1 by

$$R_1(X, Y)Z = g(Y, Z)X - g(X, Z)Y.$$

Corollary 1.7.5. *If (M^m, g) , $m \geq 2$, has constant sectional curvature k , then*

$$R(X, Y)Z = k R_1(X, Y)Z.$$

1.7.3 Ricci curvature

Definition 1.7.6 (Ricci and scalar curvature). Let (M^m, g) be a Riemannian manifold and let $\{e_1, \dots, e_m\}$ be an orthonormal basis of $T_p M$. The Ricci operator, Ricci tensor, and scalar curvature are respectively

$$\text{Ricci}(X) = \sum_{i=1}^m R(X, e_i)e_i,$$

$$\text{Ric}(X, Y) = \sum_{i=1}^m g(R(X, e_i)e_i, Y),$$

and

$$S = \text{trace}_g \text{Ric} = \sum_{i,j=1}^m g(R(e_i, e_j)e_j, e_i).$$

Remark 1.7.7. For all $X, Y \in \Gamma(TM)$,

$$\text{Ric}(X, Y) = g(\text{Ricci}(X), Y).$$

Corollary 1.7.8. *If (M^m, g) has constant sectional curvature k , then*

$$\text{Ricci}(X) = (m-1)kX,$$

$$\text{Ric}(X, Y) = (m-1)kg(X, Y),$$

and

$$S = m(m-1)k.$$

Example 1.7.9. The unit sphere S^n has constant sectional curvature $+1$, Euclidean space $\mathbb{R}^n$ has curvature 0 , and the hyperbolic plane

$$H^2 = \{(x, y) \in \mathbb{R}^2 : y > 0\}, \quad g = \frac{dx^2 + dy^2}{y^2},$$

has constant sectional curvature -1 .

1.8 Operators on Riemannian Manifolds

1.8.1 Gradient operator

Let (M, g) be a Riemannian manifold. The metric identifies covectors and vectors through the musical isomorphism

$$\sharp : \Gamma(T^*M) \longrightarrow \Gamma(TM), \quad \omega \mapsto \omega^\sharp,$$

where

$$g(\omega^\sharp, X) = \omega(X).$$

Definition 1.8.1 (Gradient). The gradient of $f \in C^\infty(M)$ is

$$\text{grad } f = (df)^\sharp.$$

Equivalently,

$$g(\text{grad } f, X) = X(f) = df(X)$$

for all $X \in \Gamma(TM)$.

In local coordinates,

$$\text{grad } f = \sum_{i,j=1}^m g^{ij} \frac{\partial f}{\partial x^i} \frac{\partial}{\partial x^j}.$$

If $\{e_i\}_{i=1}^m$ is an orthonormal frame, then

$$\text{grad } f = \sum_{i=1}^m e_i(f) e_i.$$

Proposition 1.8.2. *For $f, h \in C^\infty(M)$ and $X, Y \in \Gamma(TM)$,*

- (1) $\text{grad}(f + h) = \text{grad } f + \text{grad } h$;
- (2) $\text{grad}(fh) = h \text{grad } f + f \text{grad } h$;
- (3) $(\text{grad } f)(h) = (\text{grad } h)(f)$;
- (4) $g(\nabla_X \text{grad } f, Y) = g(\nabla_Y \text{grad } f, X)$.

1.8.2 Hessian operator

Definition 1.8.3 (Hessian). For $f \in C^\infty(M)$, the Hessian of f is the symmetric $(0, 2)$ -tensor

$$\text{Hess } f(X, Y) = g(\nabla_X \text{grad } f, Y).$$

In local coordinates,

$$\text{Hess } f = \sum_{i,j=1}^m (\text{Hess } f)_{ij} dx^i \otimes dx^j,$$

where

$$(\text{Hess } f)_{ij} = \frac{\partial^2 f}{\partial x^i \partial x^j} - \sum_{k=1}^m \Gamma_{ij}^k \frac{\partial f}{\partial x^k}.$$

1.8.3 Divergence operator

Let X be a vector field. The map

$$\nabla X : \Gamma(TM) \rightarrow \Gamma(TM), \quad Z \mapsto \nabla_Z X,$$

is a smooth bundle endomorphism.

Definition 1.8.4 (Divergence). The divergence of X is

$$\text{div } X = \text{trace}(\nabla X).$$

If $\{e_i\}_{i=1}^m$ is an orthonormal frame, then

$$\text{div } X = \sum_{i=1}^m g(\nabla_{e_i} X, e_i).$$

Property 1.8.5. For $X, Y \in \Gamma(TM)$ and $f \in C^\infty(M)$,

$$\text{div}(X + Y) = \text{div } X + \text{div } Y,$$

and

$$\text{div}(fX) = f \text{div } X + X(f).$$

Example 1.8.6 (Position field in Euclidean space). Let $P = \sum_{i=1}^n x^i \partial_i$ be the position vector field on $\mathbb{R}^n$ with the Euclidean metric. For $Y = \sum_{j=1}^n Y^j \partial_j$,

$$\nabla_Y P = \sum_{i,j=1}^n Y^j \frac{\partial x^i}{\partial x^j} \partial_i = \sum_{j=1}^n Y^j \partial_j = Y.$$

Hence

$$\text{div}_{\mathbb{R}^n} P = \sum_{i=1}^n g(\nabla_{e_i} P, e_i) = n.$$

Definition 1.8.7 (Divergence of a one-form). For $\omega \in \Gamma(T^*M)$, define

$$\text{div}_M \omega = \sum_{i=1}^m \left(e_i(\omega(e_i)) - \omega(\nabla_{e_i}^M e_i) \right).$$

Proposition 1.8.8. For $\omega, \eta \in \Gamma(T^*M)$ and $f \in C^\infty(M)$,

$$\text{div}(\omega + \eta) = \text{div } \omega + \text{div } \eta,$$

and

$$\text{div}(f\omega) = f \text{div } \omega + \omega(\text{grad } f).$$

1.8.4 Laplacian operator

Definition 1.8.9 (Laplacian). The Laplace operator on a Riemannian manifold is

$$\Delta : C^\infty(M) \longrightarrow C^\infty(M), \quad \Delta f = \operatorname{div}(\operatorname{grad} f).$$

Property 1.8.10. For $f, h \in C^\infty(M)$,

$$\Delta(f + h) = \Delta f + \Delta h,$$

and

$$\Delta(fh) = h\Delta f + f\Delta h + 2g(\operatorname{grad} f, \operatorname{grad} h).$$

1.8.5 Divergence theorem

Proposition 1.8.11 (Divergence theorem). Let D be a compact domain with boundary in a Riemannian manifold (M, g) . If ω is a one-form and X is a vector field defined near D , then

$$\int_D (\operatorname{div}_M \omega) v_D = \int_{\partial D} \omega(n) v_{\partial D},$$

and

$$\int_D (\operatorname{div}_M X) v_D = \int_{\partial D} g(X, n) v_{\partial D},$$

where n denotes the outward unit normal to ∂D .

Corollary 1.8.12. If ω is a compactly supported one-form and X is a compactly supported vector field in a domain D , then

$$\int_D \operatorname{div} \omega v_D = 0, \quad \int_D \operatorname{div} X v_D = 0.$$

1.8.6 Green's theorem

Theorem 1.8.13. Let (M, g) be a compact orientable Riemannian manifold without boundary. Then, for every $X \in \Gamma(TM)$ and every $\omega \in \Gamma(T^*M)$,

$$\int_M \operatorname{div}_M X v_g = 0, \quad \int_M \operatorname{div}_M \omega v_g = 0,$$

where in local coordinates

$$v_g = \sqrt{\det(g_{ij})} dx^1 \wedge \cdots \wedge dx^m.$$

1.9 Conformal Changes of Metric

Definition 1.9.1 (Conformal map). A smooth map $\varphi : (M, g) \rightarrow (M', g')$ is conformal if there exists a positive smooth function f such that

$$\varphi^* g' = fg.$$

Proposition 1.9.2. Let (M, g) be a Riemannian manifold of dimension m , and let

$$\tilde{g} = e^{2\gamma} g, \quad \gamma \in C^\infty(M).$$

If ∇ and $\tilde{\nabla}$ are the Levi-Civita connections of g and $\tilde{g}$, then

$$\tilde{\nabla}_X Y = \nabla_X Y + X(\gamma)Y + Y(\gamma)X - g(X, Y) \operatorname{grad} \gamma. \quad (1.4)$$

Proof. The formula follows by inserting $\tilde{g} = e^{2\gamma}g$ into the Koszul formula. Since

$$X(e^{2\gamma}) = 2e^{2\gamma}X(\gamma),$$

all extra terms collect into

$$X(\gamma)Y + Y(\gamma)X - g(X, Y) \operatorname{grad} \gamma,$$

which proves (1.4). $\square$

Proposition 1.9.3. *Under the same conformal change $\tilde{g} = e^{2\gamma}g$, the curvature tensors of $\tilde{\nabla}$ and ∇ satisfy*

$$\begin{aligned} \tilde{R}(X, Y)Z &= R(X, Y)Z + X(Y(\gamma))Z - Y(X(\gamma))X \\ &\quad + Y(\gamma)Z(\gamma)X - X(\gamma)Z(\gamma)Y - Y(\gamma)g(X, Z) \operatorname{grad} \gamma \\ &\quad + X(\gamma)g(Y, Z) \operatorname{grad} \gamma - g(Y, Z) \nabla_X \operatorname{grad} \gamma + g(X, Z) \nabla_Y \operatorname{grad} \gamma \\ &\quad - g(Y, Z) \|\operatorname{grad} \gamma\|^2 X + g(X, Z) \|\operatorname{grad} \gamma\|^2 Y. \end{aligned}$$

Proposition 1.9.4. *Under the same assumptions, the Ricci operators satisfy*

$$\widetilde{\operatorname{Ricci}}(X) = e^{-2\gamma} \left(\operatorname{Ricci}(X) - \Delta \gamma X + (2 - m) (\|\operatorname{grad} \gamma\|^2 X + \nabla_X \operatorname{grad} \gamma - X(\gamma) \operatorname{grad} \gamma) \right).$$

Chapter 2

Geometry of Submanifolds

This chapter introduces the basic notation and identities for Riemannian submanifolds: the induced connection, the normal bundle, the second fundamental form, the mean curvature vector, the Gauss equation, the shape operator, and the normal connection. The material follows standard references such as [15, 20].

2.1 Second Fundamental Form

Let M be an m -dimensional submanifold of an n -dimensional Riemannian manifold (N, h) , and let

$$i : M \hookrightarrow N$$

be the canonical inclusion. The metric induced on M is

$$g = i^*h.$$

Thus, for tangent vector fields $X, Y \in \Gamma(TM)$,

$$g(X, Y) = h(X, Y).$$

Theorem 2.1.1. *Let ∇^N be the Levi-Civita connection of (N, h) . The Levi-Civita connection ∇^M of the induced metric g on M is given by*

$$\nabla_X^M Y = (\nabla_X^N Y)^\top, \quad X, Y \in \Gamma(TM),$$

where $(\cdot)^\top : T_x N \rightarrow T_x M$ denotes the orthogonal projection.

Proof. We verify that the right-hand side defines the Levi-Civita connection on M .

First, it is a linear connection. If $X_1, X_2, Y \in \Gamma(TM)$ and $f \in C^\infty(M)$, then

$$\begin{aligned} \nabla_{fX_1+X_2}^M Y &= (\nabla_{fX_1+X_2}^N Y)^\top \\ &= (f\nabla_{X_1}^N Y + \nabla_{X_2}^N Y)^\top \\ &= f(\nabla_{X_1}^N Y)^\top + (\nabla_{X_2}^N Y)^\top \\ &= f\nabla_{X_1}^M Y + \nabla_{X_2}^M Y. \end{aligned}$$

Similarly, for $X, Y_1, Y_2 \in \Gamma(TM)$,

$$\begin{aligned} \nabla_X^M (fY_1 + Y_2) &= (\nabla_X^N (fY_1 + Y_2))^\top \\ &= (X(f)Y_1 + f\nabla_X^N Y_1 + \nabla_X^N Y_2)^\top \\ &= X(f)Y_1 + f\nabla_X^M Y_1 + \nabla_X^M Y_2. \end{aligned}$$

Second, ∇^M is compatible with g . For $X, Y, Z \in \Gamma(TM)$,

$$\begin{aligned} Xg(Y, Z) &= Xh(Y, Z) \\ &= h(\nabla_X^N Y, Z) + h(Y, \nabla_X^N Z) \\ &= h((\nabla_X^N Y)^\top, Z) + h(Y, (\nabla_X^N Z)^\top) \\ &= g(\nabla_X^M Y, Z) + g(Y, \nabla_X^M Z). \end{aligned}$$

Finally, ∇^M is torsion free. Indeed,

$$\begin{aligned} \nabla_X^M Y - \nabla_Y^M X &= (\nabla_X^N Y)^\top - (\nabla_Y^N X)^\top \\ &= (\nabla_X^N Y - \nabla_Y^N X)^\top \\ &= [X, Y]^\top = [X, Y], \end{aligned}$$

because $[X, Y]$ is tangent to M . Therefore ∇^M is the unique Levi-Civita connection of (M, g) . $\square$

Definition 2.1.2. The *normal bundle* of the submanifold (M, g) in (N, h) is the vector bundle $TM^\perp$ whose fibre at $x \in M$ is the orthogonal complement of $T_x M$ in $T_x N$:

$$T_x M^\perp = \{v \in T_x N : h(v, w) = 0 \text{ for all } w \in T_x M\}.$$

Proposition 2.1.3. Let (M, g) be a Riemannian submanifold of a Riemannian manifold (N, h) . Then, for every $x \in M$,

$$T_x N = T_x M \oplus T_x M^\perp.$$

Equivalently, every $v \in T_x N$ admits a unique decomposition

$$v = v^\top + v^\perp,$$

with $v^\top \in T_x M$ and $v^\perp \in T_x M^\perp$.

For tangent vector fields $X, Y \in \Gamma(TM)$, the ambient derivative $\nabla_X^N Y$ decomposes into tangential and normal parts:

$$\nabla_X^N Y = (\nabla_X^N Y)^\top + (\nabla_X^N Y)^\perp.$$

By Theorem 2.1.1, the tangential component is $\nabla_X^M Y$. The normal component is the second fundamental form.

Lemma 2.1.4. Let (M, g) be a Riemannian submanifold of (N, h) and define

$$B : \Gamma(TM) \times \Gamma(TM) \longrightarrow \Gamma(TM^\perp), \quad B(X, Y) = (\nabla_X^N Y)^\perp.$$

Then B is symmetric and $C^\infty(M)$ -bilinear.

Proof. Since ∇^N is torsion free,

$$\begin{aligned} B(X, Y) - B(Y, X) &= (\nabla_X^N Y)^\perp - (\nabla_Y^N X)^\perp \\ &= (\nabla_X^N Y - \nabla_Y^N X)^\perp \\ &= [X, Y]^\perp = 0, \end{aligned}$$

because $[X, Y]$ is tangent to M . Hence B is symmetric.

For $f \in C^\infty(M)$,

$$B(fX, Y) = (\nabla_{fX}^N Y)^\perp = (f\nabla_X^N Y)^\perp = fB(X, Y).$$

Using the symmetry of B gives

$$B(X, fY) = B(fY, X) = fB(Y, X) = fB(X, Y).$$

Additivity in each argument follows directly from the linearity of ∇^N . $\square$

Definition 2.1.5. The tensor field B is called the *second fundamental form* of the submanifold (M, g) in (N, h) . The decomposition

$$\nabla_X^N Y = \nabla_X^M Y + B(X, Y)$$

is the *Gauss formula*.

2.2 Curvature of a Riemannian Submanifold

Definition 2.2.1. Let (M, g) be an m -dimensional Riemannian submanifold of (N, h) , and let B be its second fundamental form. The *mean curvature vector* of M is

$$H = \frac{1}{m} \operatorname{trace}_g B = \frac{1}{m} \sum_{i=1}^m B(e_i, e_i),$$

where $\{e_1, \dots, e_m\}$ is any local orthonormal frame of TM .

Definition 2.2.2. A Riemannian submanifold M of N is called *minimal* if its mean curvature vector vanishes identically, that is,

$$H = 0.$$

Proposition 2.2.3 (Gauss equation). *Let (M, g) be a submanifold of (N, h) . Denote by R^M and R^N the curvature tensors of M and N , respectively. Then, for $X, Y, V, W \in \Gamma(TM)$,*

$$\begin{aligned} g(R^M(V, W)X, Y) &= h(R^N(V, W)X, Y) \\ &\quad - h(B(V, X), B(W, Y)) + h(B(V, Y), B(W, X)). \end{aligned}$$

Proof. By definition,

$$R^M(V, W)X = \nabla_V^M \nabla_W^M X - \nabla_W^M \nabla_V^M X - \nabla_{[V, W]}^M X.$$

Hence

$$\begin{aligned} g(R^M(V, W)X, Y) &= g(\nabla_V^M \nabla_W^M X, Y) - g(\nabla_W^M \nabla_V^M X, Y) \\ &\quad - g(\nabla_{[V, W]}^M X, Y). \end{aligned}$$

We compute the first term. Since $\nabla_W^M X = (\nabla_W^N X)^\top = \nabla_W^N X - B(W, X)$,

$$\begin{aligned} g(\nabla_V^M \nabla_W^M X, Y) &= h(\nabla_V^N \nabla_W^M X, Y) \\ &= h(\nabla_V^N (\nabla_W^N X - B(W, X)), Y) \\ &= h(\nabla_V^N \nabla_W^N X, Y) - h(\nabla_V^N B(W, X), Y). \end{aligned}$$

Now $B(W, X)$ is normal and Y is tangent, so $h(B(W, X), Y) = 0$. Differentiating this identity in the direction V gives

$$h(\nabla_V^N B(W, X), Y) + h(B(W, X), \nabla_V^N Y) = 0.$$

Therefore,

$$-h(\nabla_V^N B(W, X), Y) = h(B(W, X), \nabla_V^N Y).$$

Only the normal component of $\nabla_V^N Y$ contributes, so

$$h(B(W, X), \nabla_V^N Y) = h(B(W, X), B(V, Y)).$$

Thus

$$g(\nabla_V^M \nabla_W^M X, Y) = h(\nabla_V^N \nabla_W^N X, Y) + h(B(W, X), B(V, Y)).$$

Similarly,

$$g(\nabla_W^M \nabla_V^M X, Y) = h(\nabla_W^N \nabla_V^N X, Y) + h(B(V, X), B(W, Y)),$$

and

$$g(\nabla_{[V,W]}^M X, Y) = h(\nabla_{[V,W]}^N X, Y).$$

Combining these identities yields the desired formula. $\square$

Corollary 2.2.4. *Let (M, g) be a Riemannian submanifold of (N, h) . If K_M and K_N denote the sectional curvatures of M and N , respectively, then for any two linearly independent vectors $v, w \in T_x M$,*

$$K_M(v, w) = K_N(v, w) + \frac{h(B(v, v), B(w, w)) - h(B(v, w), B(v, w))}{g(v, v)g(w, w) - g(v, w)^2}.$$

Example 2.2.5. Let

$$M = S^n(r) = \{x \in \mathbb{R}^{n+1} : \|x\| = r\},$$

and let $N = \mathbb{R}^{n+1}$ with its Euclidean metric. Denote by

$$P = \sum_{i=1}^{n+1} x_i \partial_i$$

the position vector field. For every vector field X on $\mathbb{R}^{n+1}$,

$$\nabla_X^{\mathbb{R}^{n+1}} P = X.$$

Along $S^n(r)$, the position field P is normal to the sphere. Thus, for tangent vector fields $X, Y \in \Gamma(TS^n(r))$, the second fundamental form has the form

$$B(X, Y) = \alpha P$$

for some function $\alpha \in C^\infty(M)$. Taking the inner product with P gives

$$\begin{aligned} h(B(X, Y), P) &= \alpha h(P, P) = \alpha r^2, \\ h(B(X, Y), P) &= h((\nabla_X^{\mathbb{R}^{n+1}} Y)^\perp, P) = h(\nabla_X^{\mathbb{R}^{n+1}} Y, P) \\ &= Xh(Y, P) - h(Y, \nabla_X^{\mathbb{R}^{n+1}} P) = -h(Y, X). \end{aligned}$$

Therefore

$$\alpha = -\frac{1}{r^2} h(X, Y),$$

and hence

$$B(X, Y) = -\frac{1}{r^2} h(X, Y) P.$$

Since the ambient Euclidean curvature is zero, Corollary 2.2.4 gives

$$K_{S^n(r)}(v, w) = \frac{1}{r^2}.$$

Thus the round sphere of radius r has constant sectional curvature $1/r^2$.

2.3 Shape Operator

Definition 2.3.1. Let (M, g) be a Riemannian submanifold of (N, h) and let U be a normal vector field along M . The *shape operator*, also called the *Weingarten map*, associated with U is

$$A_U : \Gamma(TM) \longrightarrow \Gamma(TM), \quad A_U X = -(\nabla_X^N U)^\top.$$

Proposition 2.3.2. Let (M, g) be a Riemannian submanifold of (N, h) , and let U be a normal vector field along M . Then

$$g(A_U X, Y) = h(B(X, Y), U), \quad X, Y \in \Gamma(TM).$$

Proof. For $X, Y \in \Gamma(TM)$,

$$\begin{aligned} g(A_U X, Y) &= -h((\nabla_X^N U)^\top, Y) \\ &= -h(\nabla_X^N U, Y) \\ &= -Xh(U, Y) + h(U, \nabla_X^N Y) \\ &= h(U, (\nabla_X^N Y)^\perp) \\ &= h(U, B(X, Y)). \end{aligned}$$

Since h is symmetric, this is the desired identity. $\square$

Example 2.3.3. Consider the cylinder

$$C = \{(x, y, z) \in \mathbb{R}^3 : x^2 + y^2 = r^2, -1 < z < 1\}, \quad r > 0.$$

A parametrization is

$$(\theta, z) \longmapsto (r \cos \theta, r \sin \theta, z).$$

Thus

$$\partial_\theta = (-r \sin \theta, r \cos \theta, 0), \quad \partial_z = (0, 0, 1).$$

Choose the unit normal field

$$\eta = (-\cos \theta, -\sin \theta, 0).$$

Then

$$\begin{aligned} A_\eta(\partial_\theta) &= -(\nabla_{\partial_\theta}^{\mathbb{R}^3} \eta)^\top = -\nabla_{\partial_\theta}^{\mathbb{R}^3} \eta \\ &= (-\sin \theta, \cos \theta, 0) = \frac{1}{r} \partial_\theta, \end{aligned}$$

and, by the same calculation,

$$A_\eta(\partial_z) = 0.$$

With respect to the frame $\{\partial_\theta, \partial_z\}$, the shape operator is represented by

$$A_\eta = \begin{pmatrix} \frac{1}{r} & 0 \\ 0 & 0 \end{pmatrix}.$$

Example 2.3.4. Let U be a normal vector field along $S^n(r) \subset \mathbb{R}^{n+1}$. From Example 2.2.5,

$$B(X, Y) = -\frac{1}{r^2} g(X, Y) P.$$

Using Proposition 2.3.2, we obtain

$$g(A_U X, Y) = h(B(X, Y), U) = -\frac{1}{r^2} g(X, Y) h(P, U).$$

Consequently,

$$A_U X = -\frac{1}{r^2} h(P, U) X, \quad X \in \Gamma(TS^n(r)).$$

Definition 2.3.5. Let (M, g) be a Riemannian submanifold of (N, h) . The *normal connection* of M in N is the map

$$\nabla^\perp : \Gamma(TM) \times \Gamma(TM^\perp) \longrightarrow \Gamma(TM^\perp), \quad \nabla_X^\perp Y = (\nabla_X^N Y)^\perp.$$

Property 2.3.6. Let (M, g) be a Riemannian submanifold of (N, h) . The normal connection satisfies:

i) $\nabla_X^\perp Y$ is $C^\infty(M)$ -linear in X and $\mathbb{R}$ -linear in Y ;

ii)

$$\nabla_X^\perp(fY) = X(f)Y + f\nabla_X^\perp Y,$$

for $X \in \Gamma(TM)$, $Y \in \Gamma(TM^\perp)$, and $f \in C^\infty(M)$;

iii)

$$Xh(Y, Z) = h(\nabla_X^\perp Y, Z) + h(Y, \nabla_X^\perp Z),$$

for $X \in \Gamma(TM)$ and $Y, Z \in \Gamma(TM^\perp)$.

Proof. These identities follow directly from the definition of $\nabla^\perp$ and the corresponding properties of the ambient Levi-Civita connection ∇^N . $\square$

Chapter 3

Harmonic and Biharmonic Mappings

In this chapter we introduce the energy functional for smooth maps between Riemannian manifolds and recall the variational equations that characterize harmonic maps. We then pass to the bienergy functional, biharmonic maps, and the associated stress-energy tensors.

3.1 Harmonic Maps

Definition 3.1.1. Let (M^m, g) and (N^n, h) be Riemannian manifolds, and let

$$\varphi \in C^\infty(M, N).$$

For a compact domain $D \subset M$, the energy of φ on D is

$$E(\varphi) = \frac{1}{2} \int_D |d\varphi|^2 v_g, \quad (3.1)$$

where $|d\varphi|$ denotes the Hilbert-Schmidt norm of the differential $d\varphi$, and v_g is the volume element of (M, g) . In local coordinates,

$$v_g = \sqrt{\det(g_{ij})} dx^1 \wedge \cdots \wedge dx^m.$$

Remark 3.1.2. If $\{e_1, \dots, e_m\}$ is a local orthonormal frame on (M, g) , then

$$|d\varphi|^2 = \sum_{i=1}^m h(d\varphi(e_i), d\varphi(e_i)).$$

Equivalently, if $(x^1, \dots, x^m)$ are local coordinates on M and $(y^1, \dots, y^n)$ are local coordinates on N , then

$$\begin{aligned} |d\varphi|^2 &= \sum_{i=1}^m h(d\varphi(e_i), d\varphi(e_i)) \\ &= \sum_{a,b=1}^m \sum_{\alpha,\beta=1}^n g^{ab} \frac{\partial \varphi^\alpha}{\partial x^a} \frac{\partial \varphi^\beta}{\partial x^b} (h_{\alpha\beta} \circ \varphi). \end{aligned}$$

Example 3.1.3. Consider the map

$$\varphi : \mathbb{R}^n \setminus \{0\} \longrightarrow \mathbb{R}^n \setminus \{0\}, \quad x \longmapsto \frac{x}{\|x\|^\ell}.$$

Writing $x = (x^1, \dots, x^n)$, one has

$$d\varphi \left(\frac{\partial}{\partial x^i} \right) = \|x\|^{-\ell} \left(\frac{\partial}{\partial y^i} \circ \varphi \right) - \ell x^i \|x\|^{-\ell-2} \sum_{\alpha=1}^n x^\alpha \left(\frac{\partial}{\partial y^\alpha} \circ \varphi \right).$$

Therefore,

$$\begin{aligned} |d\varphi|^2 &= \sum_{i=1}^n \left\langle \|x\|^{-\ell} \frac{\partial}{\partial y^i} - \ell x^i \|x\|^{-\ell-2} \sum_{\alpha=1}^n x^\alpha \frac{\partial}{\partial y^\alpha}, \|x\|^{-\ell} \frac{\partial}{\partial y^i} - \ell x^i \|x\|^{-\ell-2} \sum_{\beta=1}^n x^\beta \frac{\partial}{\partial y^\beta} \right\rangle \circ \varphi \\ &= n\|x\|^{-2\ell} - 2\ell\|x\|^{-2\ell} + \ell^2\|x\|^{-2\ell} \\ &= (n - 2\ell + \ell^2)\|x\|^{-2\ell}. \end{aligned}$$

Definition 3.1.4. A variation of $\varphi \in C^\infty(M, N)$ supported in a compact domain $D \subset M$ is a smooth family of maps

$$(\varphi_t)_{t \in (-\varepsilon, \varepsilon)} : M \longrightarrow N$$

with $\varphi_0 = \varphi$ and $\varphi_t = \varphi$ on $M \setminus \text{int}(D)$.

Definition 3.1.5. A smooth map $\varphi : (M, g) \rightarrow (N, h)$ is called harmonic if it is a critical point of the energy functional on every compact domain $D \subset M$, that is,

$$\left. \frac{d}{dt} E(\varphi_t) \right|_{t=0} = 0$$

for every compactly supported variation (φ_t) of φ .

3.1.1 First Variation of Energy

Definition 3.1.6. Let $\varphi : (M, g) \rightarrow (N, h)$ be a smooth map. The trace of the second fundamental form of φ is called the tension field of φ and is denoted by

$$\tau(\varphi) = \text{trace}_g \nabla d\varphi. \quad (3.2)$$

In local coordinates, the tension field is given by

$$\tau(\varphi) = \sum_{i,j=1}^m \sum_{\gamma=1}^n g^{ij} \left(\frac{\partial^2 \varphi^\gamma}{\partial x^i \partial x^j} + \sum_{\alpha, \beta=1}^n \frac{\partial \varphi^\alpha}{\partial x^i} \frac{\partial \varphi^\beta}{\partial x^j} ({}^N \Gamma_{\alpha\beta}^\gamma \circ \varphi) - \sum_{k=1}^m {}^M \Gamma_{ij}^k \frac{\partial \varphi^\gamma}{\partial x^k} \right) \left(\frac{\partial}{\partial y^\gamma} \circ \varphi \right). \quad (3.3)$$

Here $\partial/\partial x^i$ and $\partial/\partial y^\alpha$ denote the coordinate vector fields on M and N , respectively.

Theorem 3.1.7 (First variation of energy). *Let $\varphi : (M^m, g) \rightarrow (N^n, h)$ be a smooth map, and let $(\varphi_t)_{t \in (-\varepsilon, \varepsilon)}$ be a smooth variation supported in D . If*

$$v = \left. \frac{d\varphi_t}{dt} \right|_{t=0}$$

is the corresponding variation vector field, then

$$\left. \frac{d}{dt} E(\varphi_t) \right|_{t=0} = - \int_D h(v, \tau(\varphi)) v_g. \quad (3.4)$$

Proof. Let $\{e_1, \dots, e_m\}$ be a local orthonormal frame on (M, g) , and define

$$\Phi : M \times (-\varepsilon, \varepsilon) \longrightarrow N, \quad \Phi(x, t) = \varphi_t(x).$$

At $t = 0$ we have

$$d\Phi(e_i, 0) = d\varphi(e_i), \quad d\Phi\left(0, \frac{d}{dt}\right) = v.$$

Since the pull-back connection is torsion free along the variation and $[(e_i, 0), (0, d/dt)] = 0$, one obtains

$$\begin{aligned}
\left. \frac{d}{dt} E(\varphi_t) \right|_{t=0} &= \frac{1}{2} \int_D \left. \frac{\partial}{\partial t} |d\varphi_t|^2 \right|_{t=0} v_g \\
&= \int_D \sum_{i=1}^m h(\nabla_{(0, d/dt)}^\Phi d\Phi(e_i, 0), d\Phi(e_i, 0)) \Big|_{t=0} v_g \\
&= \int_D \sum_{i=1}^m h(\nabla_{e_i}^\varphi v, d\varphi(e_i)) v_g \\
&= \int_D \sum_{i=1}^m \{e_i h(v, d\varphi(e_i)) - h(v, \nabla_{e_i}^\varphi d\varphi(e_i))\} v_g.
\end{aligned}$$

Define a compactly supported one-form ω on D by

$$\omega(X) = h(v, d\varphi(X)), \quad X \in \Gamma(TM).$$

Then

$$\begin{aligned}
\operatorname{div}_M \omega &= \sum_{i=1}^m (\nabla_{e_i} \omega)(e_i) \\
&= \sum_{i=1}^m \{e_i(\omega(e_i)) - \omega(\nabla_{e_i}^M e_i)\} \\
&= \sum_{i=1}^m \{e_i h(v, d\varphi(e_i)) - h(v, d\varphi(\nabla_{e_i}^M e_i))\}.
\end{aligned}$$

Using the definition of $\tau(\varphi)$, this gives

$$\left. \frac{d}{dt} E(\varphi_t) \right|_{t=0} = \int_D (\operatorname{div}_M \omega) v_g - \int_D h(v, \tau(\varphi)) v_g.$$

The divergence term vanishes because the variation is supported in D . Hence (3.4) follows. $\square$

Theorem 3.1.8. *A smooth map $\varphi \in C^\infty(M, N)$ between two Riemannian manifolds is harmonic if and only if*

$$\tau(\varphi) = \operatorname{trace}_g \nabla d\varphi = 0.$$

3.1.2 Second Variation of Energy

Theorem 3.1.9 (Second variation of energy). *Let $\varphi : (M^m, g) \rightarrow (N^n, h)$ be a harmonic map, and let $D \subset M$ be a compact domain. If $(\varphi_{t,s})$ is a two-parameter variation of φ with compact support in D , then*

$$\left. \frac{\partial^2}{\partial t \partial s} E(\varphi_{t,s}) \right|_{(t,s)=(0,0)} = \int_D h(J_\varphi(V), W) v_g, \quad (3.5)$$

where

$$V = \left. \frac{\partial \varphi_{t,s}}{\partial t} \right|_{(0,0)}, \quad W = \left. \frac{\partial \varphi_{t,s}}{\partial s} \right|_{(0,0)},$$

and

$$J_\varphi(V) = -\operatorname{trace}_g R^N(V, d\varphi)d\varphi - \operatorname{trace}_g (\nabla^\varphi)^2 V. \quad (3.6)$$

Proof. Let $\{e_1, \dots, e_m\}$ be an orthonormal frame on (M, g) and set

$$\Phi : M \times (-\varepsilon, \varepsilon) \times (-\varepsilon, \varepsilon) \longrightarrow N, \quad \Phi(x, t, s) = \varphi_{t,s}(x).$$

Write

$$E_i = (e_i, 0, 0), \quad \partial_t = (0, d/dt, 0), \quad \partial_s = (0, 0, d/ds).$$

A direct differentiation gives

$$\left. \frac{\partial^2}{\partial t \partial s} E(\varphi_{t,s}) \right|_{(0,0)} = \frac{1}{2} \int_D \sum_{i=1}^m \left. \frac{\partial^2}{\partial t \partial s} h(d\Phi(E_i), d\Phi(E_i)) \right|_{(0,0)} v_g. \quad (3.7)$$

Furthermore,

$$\begin{aligned} \frac{1}{2} \frac{\partial^2}{\partial t \partial s} h(d\Phi(E_i), d\Phi(E_i)) &= h(\nabla_{\partial_t}^\Phi \nabla_{\partial_s}^\Phi d\Phi(E_i), d\Phi(E_i)) \\ &\quad + h(\nabla_{\partial_s}^\Phi d\Phi(E_i), \nabla_{\partial_t}^\Phi d\Phi(E_i)). \end{aligned} \quad (3.8)$$

The curvature identity and $[\partial_t, E_i] = 0$ yield

$$\nabla_{\partial_t}^\Phi \nabla_{\partial_s}^\Phi d\Phi(E_i) = R^N(d\Phi(\partial_t), d\Phi(E_i))d\Phi(\partial_s) + \nabla_{E_i}^\Phi \nabla_{\partial_t}^\Phi d\Phi(\partial_s). \quad (3.9)$$

Define a one-form ω on M by

$$\omega(X) = h\left(\nabla_{\partial_t}^\Phi d\Phi(\partial_s) \Big|_{(0,0)}, d\varphi(X)\right).$$

Since φ is harmonic, $\tau(\varphi) = 0$, and therefore

$$\operatorname{div}_M \omega = \sum_{i=1}^m h\left(\nabla_{E_i}^\Phi \nabla_{\partial_t}^\Phi d\Phi(\partial_s) \Big|_{(0,0)}, d\varphi(e_i)\right). \quad (3.10)$$

Thus, after evaluating at $(0, 0)$,

$$\sum_{i=1}^m h(\nabla_{\partial_t}^\Phi \nabla_{\partial_s}^\Phi d\Phi(E_i), d\Phi(E_i)) = \sum_{i=1}^m h(R^N(V, d\varphi(e_i))W, d\varphi(e_i)) + \operatorname{div}_M \omega. \quad (3.11)$$

For the second term in (3.8), define

$$\eta(X) = h(W, \nabla_X^\varphi V).$$

Then

$$\operatorname{div}_M \eta = \sum_{i=1}^m \{e_i h(W, \nabla_{e_i}^\varphi V) - h(W, \nabla_{\nabla_{e_i}^M V}^\varphi V)\}. \quad (3.12)$$

Using the product rule, one obtains

$$\sum_{i=1}^m h(\nabla_{\partial_t}^\Phi d\Phi(E_i), \nabla_{\partial_s}^\Phi d\Phi(E_i)) \Big|_{(0,0)} = \operatorname{div}_M \eta + \sum_{i=1}^m h(W, \nabla_{\nabla_{e_i}^M V}^\varphi V) - \sum_{i=1}^m h(W, \nabla_{e_i}^\varphi \nabla_{e_i}^\varphi V). \quad (3.13)$$

Combining (3.7), (3.11), and (3.13), and applying the divergence theorem, gives

$$\begin{aligned} \left. \frac{\partial^2}{\partial t \partial s} E(\varphi_{t,s}) \right|_{(0,0)} &= \int_D \sum_{i=1}^m \left\{ -h(R^N(V, d\varphi(e_i))d\varphi(e_i), W) \right. \\ &\quad \left. + h(W, \nabla_{\nabla_{e_i}^M V}^\varphi V) - h(W, \nabla_{e_i}^\varphi \nabla_{e_i}^\varphi V) \right\} v_g \\ &= \int_D h(J_\varphi(V), W) v_g. \end{aligned}$$

□

Definition 3.1.10. Let $\varphi : (M^m, g) \rightarrow (N^n, h)$ be a smooth map between Riemannian manifolds. The Jacobi operator of φ is the operator

$$J_\varphi : \Gamma(\varphi^{-1}TN) \longrightarrow \Gamma(\varphi^{-1}TN)$$

defined by

$$\begin{aligned} J_\varphi(v) &= -\text{trace}_g R^N(v, d\varphi)d\varphi - \text{trace}_g(\nabla^\varphi)^2 v \\ &= -\sum_{i=1}^m R^N(v, d\varphi(e_i))d\varphi(e_i) - \sum_{i=1}^m \left(\nabla_{e_i}^\varphi \nabla_{e_i}^\varphi v - \nabla_{\nabla_{e_i}^M e_i}^\varphi v \right), \end{aligned} \quad (3.14)$$

where $\{e_1, \dots, e_m\}$ is a local orthonormal frame on (M, g) .

Property 3.1.11. Let $\varphi : (M^m, g) \rightarrow (N^n, h)$ be a smooth map. Then:

- (i) J_φ is $\mathbb{R}$ -linear;
- (ii) for every $v \in \Gamma(\varphi^{-1}TN)$ and $f \in C^\infty(M)$,

$$J_\varphi(fv) = fJ_\varphi(v) - \left(\Delta f v + 2\nabla_{\text{grad } f}^\varphi v \right), \quad (3.15)$$

where $\Delta f = \text{div}(\text{grad } f)$.

Proof. Linearity follows directly from the linearity of the curvature tensor and of the pull-back connection. For the product rule, one expands

$$\begin{aligned} J_\varphi(fv) &= -\sum_{i=1}^m \left\{ R^N(fv, d\varphi(e_i))d\varphi(e_i) + \nabla_{e_i}^\varphi \nabla_{e_i}^\varphi(fv) - \nabla_{\nabla_{e_i}^M e_i}^\varphi(fv) \right\} \\ &= fJ_\varphi(v) - \left(\sum_{i=1}^m \{e_i(e_i f) - (\nabla_{e_i}^M e_i)f\} \right) v - 2 \sum_{i=1}^m e_i(f) \nabla_{e_i}^\varphi v, \end{aligned}$$

which is exactly (3.15). □

3.2 Biharmonic Maps

Definition 3.2.1. Let $\varphi : (M, g) \rightarrow (N, h)$ be a smooth map and let $D \subset M$ be a compact domain. The bienergy of φ on D is

$$E_2(\varphi) = \frac{1}{2} \int_D |\tau(\varphi)|^2 v_g, \quad (3.16)$$

where $|\tau(\varphi)|^2 = h(\tau(\varphi), \tau(\varphi))$.

Definition 3.2.2. A smooth map $\varphi : (M^m, g) \rightarrow (N^n, h)$ is called biharmonic if it is a critical point of the bienergy functional on every compact domain $D \subset M$, that is,

$$\left. \frac{d}{dt} E_2(\varphi_t) \right|_{t=0} = 0 \quad (3.17)$$

for every compactly supported variation (φ_t) of φ .

3.2.1 First Variation of Bienergy

Theorem 3.2.3 (First variation of bienergy). *Let $\varphi : (M^m, g) \rightarrow (N^n, h)$ be a smooth map, and let $(\varphi_t)_{t \in (-\varepsilon, \varepsilon)}$ be a smooth variation supported in D . Then*

$$\left. \frac{d}{dt} E_2(\varphi_t) \right|_{t=0} = - \int_D h(v, \tau_2(\varphi)) v_g, \quad (3.18)$$

where $v = d\varphi_t/dt|_{t=0}$ is the variation vector field and

$$\begin{aligned} \tau_2(\varphi) &= -\text{trace}_g R^N(\tau(\varphi), d\varphi) d\varphi - \text{trace}_g (\nabla^\varphi)^2 \tau(\varphi) \\ &= -\sum_{i=1}^m R^N(\tau(\varphi), d\varphi(e_i)) d\varphi(e_i) - \sum_{i=1}^m \left\{ \nabla_{e_i}^\varphi \nabla_{e_i}^\varphi \tau(\varphi) - \nabla_{\nabla_{e_i}^M e_i}^\varphi \tau(\varphi) \right\}. \end{aligned} \quad (3.19)$$

Proof. Let

$$\Phi : M \times (-\varepsilon, \varepsilon) \longrightarrow N, \quad \Phi(x, t) = \varphi_t(x).$$

Then

$$\left. \frac{d}{dt} E_2(\varphi_t) \right|_{t=0} = \int_D \sum_{i,j=1}^m h \left(\nabla_{(0, d/dt)}^\Phi \nabla d\Phi((e_i, 0), (e_i, 0)) \Big|_{t=0}, \nabla d\varphi(e_j, e_j) \right) v_g. \quad (3.20)$$

Since $[(0, d/dt), (e_i, 0)] = 0$, we have

$$\nabla_{(0, d/dt)}^\Phi d\Phi(e_i, 0) = \nabla_{(e_i, 0)}^\Phi d\Phi(0, d/dt),$$

and similarly

$$\nabla_{(0, d/dt)}^\Phi d\Phi(\nabla_{e_i}^M e_i, 0) = \nabla_{(\nabla_{e_i}^M e_i, 0)}^\Phi d\Phi(0, d/dt).$$

Thus

$$\nabla_{(0, d/dt)}^\Phi \nabla d\Phi((e_i, 0), (e_i, 0)) \Big|_{t=0} = R^N(v, d\varphi(e_i)) d\varphi(e_i) + \nabla_{e_i}^\varphi \nabla_{e_i}^\varphi v - \nabla_{\nabla_{e_i}^M e_i}^\varphi v. \quad (3.21)$$

Let w be the one-form supported in D defined by

$$w(X) = h(\nabla_X^\varphi v, \tau(\varphi)).$$

Then

$$\text{div}_M w = \sum_{i=1}^m \{ h(\nabla_{e_i}^\varphi \nabla_{e_i}^\varphi v, \tau(\varphi)) + h(\nabla_{e_i}^\varphi v, \nabla_{e_i}^\varphi \tau(\varphi)) - h(\nabla_{\nabla_{e_i}^M e_i}^\varphi v, \tau(\varphi)) \}. \quad (3.22)$$

Similarly, define

$$\eta(X) = h(v, \nabla_X^\varphi \tau(\varphi)).$$

Then

$$\text{div}_M \eta = \sum_{i=1}^m \{ h(\nabla_{e_i}^\varphi v, \nabla_{e_i}^\varphi \tau(\varphi)) + h(v, \nabla_{e_i}^\varphi \nabla_{e_i}^\varphi \tau(\varphi)) - h(v, \nabla_{\nabla_{e_i}^M e_i}^\varphi \tau(\varphi)) \}. \quad (3.23)$$

Combining (3.20), (3.21), (3.22), and (3.23), and using the divergence theorem, gives

$$\begin{aligned} \left. \frac{d}{dt} E_2(\varphi_t) \right|_{t=0} &= - \int_D h \left(v, -\sum_{i=1}^m R^N(\tau(\varphi), d\varphi(e_i)) d\varphi(e_i) - \sum_{i=1}^m \{ \nabla_{e_i}^\varphi \nabla_{e_i}^\varphi \tau(\varphi) - \nabla_{\nabla_{e_i}^M e_i}^\varphi \tau(\varphi) \} \right) v_g \\ &= - \int_D h(v, \tau_2(\varphi)) v_g. \end{aligned}$$

□

Theorem 3.2.4. *A smooth map $\varphi \in C^\infty(M, N)$ between two Riemannian manifolds is biharmonic if and only if*

$$\tau_2(\varphi) = -\text{trace}_g R^N(\tau(\varphi), d\varphi)d\varphi - \text{trace}_g(\nabla^\varphi)^2 \tau(\varphi) = 0. \quad (3.24)$$

Remark 3.2.5. Equation (3.24) is the Euler-Lagrange equation associated with the bienergy functional. In the notation of the Jacobi operator,

$$\tau_2(\varphi) = J_\varphi(\tau(\varphi)).$$

3.3 Stress-Energy Tensor

Proposition 3.3.1. *Let $\varphi : (M^m, g) \rightarrow (N^n, h)$ be a smooth map, let $D \subset M$ be a compact domain, and let $(g_t)_{t \in (-\varepsilon, \varepsilon)}$ be a variation of the metric g supported in D , with $g_0 = g$. Then*

$$\delta g = \left. \frac{\partial g_t}{\partial t} \right|_{t=0} \in \Gamma(T^*M \otimes T^*M).$$

Locally, if $g_t = g_{ij}(t, x)dx^i \otimes dx^j$, then

$$\delta g = \left. \frac{\partial g_{ij}}{\partial t} \right|_{t=0} dx^i \otimes dx^j.$$

Definition 3.3.2. Let $\varphi : (M^m, g) \rightarrow (N^n, h)$ be a smooth map. The stress-energy tensor associated with φ is

$$S(\varphi) = e(\varphi)g - \varphi^*h, \quad (3.25)$$

where

$$e(\varphi) = \frac{1}{2}|d\varphi|^2$$

is the energy density of φ . Equivalently, for $X, Y \in \Gamma(TM)$,

$$S(\varphi)(X, Y) = e(\varphi)g(X, Y) - h(d\varphi(X), d\varphi(Y)). \quad (3.26)$$

Proposition 3.3.3. *Let $\varphi : (M^m, g) \rightarrow (N^n, h)$ be a smooth map, let $D \subset M$ be a compact domain, and let (g_t) be a smooth variation of the metric g . Then*

$$\left. \frac{d}{dt} E(\varphi) \right|_{t=0} = - \int_D \langle S(\varphi), \delta g \rangle v_g. \quad (3.27)$$

Theorem 3.3.4. *Let $\varphi : (M^m, g) \rightarrow (N^n, h)$ be a smooth map. Then*

$$\text{div}_M S(\varphi) = -h(\tau(\varphi), d\varphi). \quad (3.28)$$

More precisely, as a one-form,

$$(\text{div}_M S(\varphi))(X) = -h(\tau(\varphi), d\varphi(X)).$$

Proof. Let $\{e_1, \dots, e_m\}$ be a local orthonormal frame such that $\nabla_{e_i}^M e_j = 0$ at a fixed point $x \in M$. For $X = e_j$ at x , we compute

$$\begin{aligned} (\text{div}_M S(\varphi))(e_j) &= \sum_{i=1}^m e_i(S(\varphi)(e_i, e_j)) \\ &= \frac{1}{2} \sum_{i=1}^m e_i(|d\varphi|^2) \delta_{ij} - \sum_{i=1}^m e_i h(d\varphi(e_i), d\varphi(e_j)) \\ &= \sum_{k=1}^m h(\nabla_{e_j}^\varphi d\varphi(e_k), d\varphi(e_k)) - h\left(\sum_{i=1}^m \nabla_{e_i}^\varphi d\varphi(e_i), d\varphi(e_j)\right) \\ &\quad - \sum_{i=1}^m h(d\varphi(e_i), \nabla_{e_i}^\varphi d\varphi(e_j)). \end{aligned}$$

Since $\nabla d\varphi$ is symmetric at the point under consideration, the first and third sums cancel. Hence

$$(\operatorname{div}_M S(\varphi))(e_j) = -h(\tau(\varphi), d\varphi(e_j)),$$

which proves the formula. $\square$

Corollary 3.3.5. *Let $\varphi : (M^m, g) \rightarrow (N^n, h)$ be a smooth map between Riemannian manifolds.*

- (i) *If φ is harmonic, then $\operatorname{div}_M S(\varphi) = 0$.*
- (ii) *If φ is a submersion and $\operatorname{div}_M S(\varphi) = 0$, then φ is harmonic.*

3.4 Stress Bienergy Tensor

Definition 3.4.1. Let $\varphi : (M^m, g) \rightarrow (N^n, h)$ be a smooth map. The stress bienergy tensor associated with φ is the symmetric $(0, 2)$ -tensor $S_2(\varphi)$ defined by

$$\begin{aligned} S_2(\varphi)(X, Y) = & -\frac{1}{2}|\tau(\varphi)|^2 g(X, Y) - \langle d\varphi, \nabla^\varphi \tau(\varphi) \rangle g(X, Y) \\ & + h(d\varphi(X), \nabla_Y^\varphi \tau(\varphi)) + h(d\varphi(Y), \nabla_X^\varphi \tau(\varphi)), \end{aligned} \quad (3.29)$$

where

$$\langle d\varphi, \nabla^\varphi \tau(\varphi) \rangle = \sum_{i=1}^m h(d\varphi(e_i), \nabla_{e_i}^\varphi \tau(\varphi))$$

with respect to a local orthonormal frame $\{e_1, \dots, e_m\}$ on (M, g) .

Proposition 3.4.2. *Let $\varphi : (M^m, g) \rightarrow (N^n, h)$ be a smooth map, let $D \subset M$ be a compact domain, and let (g_t) be a smooth variation of the metric g . Then*

$$\left. \frac{d}{dt} E_2(\varphi) \right|_{t=0} = - \int_D \langle S_2(\varphi), \delta g \rangle v_g. \quad (3.30)$$

The proof uses two elementary identities for the metric variation. We write

$$\xi = (\operatorname{div}_M(\delta g))^\sharp - \frac{1}{2} \operatorname{grad}_M(\operatorname{trace}_g \delta g),$$

where $\sharp$ denotes the musical isomorphism induced by g .

Lemma 3.4.3. *Let $\varphi : (M^m, g) \rightarrow (N^n, h)$ be a smooth map and let (g_t) be a smooth variation of the metric. Then*

$$\delta(|\tau(\varphi)|^2) = -2 \langle h(\nabla d\varphi, \tau(\varphi)), \delta g \rangle - 2h(d\varphi(\xi), \tau(\varphi)). \quad (3.31)$$

Proof. In local coordinates, the variation of the tension field satisfies

$$\delta(\tau(\varphi)^\alpha) = -g^{ai} g^{bj} \delta(g_{ab}) (\nabla d\varphi)_{ij}^\alpha - \xi^k \varphi_k^\alpha.$$

Multiplying by $2\tau(\varphi)^\beta h_{\alpha\beta}$ gives exactly (3.31). $\square$

Lemma 3.4.4. *Let $\omega = h(d\varphi, \tau(\varphi))$. Then*

$$\int_D h(d\varphi(\xi), \tau(\varphi)) v_g = \int_D \left\langle -\operatorname{sym}(\nabla \omega) + \frac{1}{2} \operatorname{div}_M(\omega^\sharp) g, \delta g \right\rangle v_g. \quad (3.32)$$

Proof. Since $h(d\varphi(\xi), \tau(\varphi)) = \omega(\xi)$, we have

$$\int_D \omega(\xi) v_g = \int_D \omega((\operatorname{div}_M \delta g)^\sharp) v_g - \frac{1}{2} \int_D \omega(\operatorname{grad}_M(\operatorname{trace}_g \delta g)) v_g.$$

For a symmetric tensor σ , one uses the identity

$$g^*(\omega, \operatorname{div}_M \sigma) = \operatorname{div}_M(C(\omega, \sigma)^\sharp) - \langle \operatorname{sym}(\nabla \omega), \sigma \rangle,$$

where $C(\omega, \sigma) = \omega_i \sigma_{ij} dx^j$. Taking $\sigma = \delta g$ and integrating eliminates the divergence term. Also,

$$\omega(\operatorname{grad}_M \lambda) = g^*(\omega, d\lambda),$$

with $\lambda = \operatorname{trace}_g \delta g$, and integration by parts gives

$$-\frac{1}{2} \int_D \omega(\operatorname{grad}_M(\operatorname{trace}_g \delta g)) v_g = \frac{1}{2} \int_D \langle \operatorname{div}_M(\omega^\sharp) g, \delta g \rangle v_g.$$

Combining these identities proves (3.32). $\square$

Proof of Proposition 3.4.2. Using Lemma 3.4.3, the variation of the volume form, and Lemma 3.4.4, we obtain

$$\begin{aligned} \left. \frac{d}{dt} E_2(\varphi) \right|_{t=0} &= \frac{1}{2} \int_D \delta(|\tau(\varphi)|^2) v_g + \frac{1}{2} \int_D |\tau(\varphi)|^2 \delta(v_g) \\ &= \frac{1}{2} \int_D \left\langle -2h(\nabla d\varphi, \tau(\varphi)) + 2\operatorname{sym}(\nabla h(d\varphi, \tau(\varphi))) \right. \\ &\quad \left. - \operatorname{div}_M(h(d\varphi, \tau(\varphi))^\sharp) g + \frac{1}{2} |\tau(\varphi)|^2 g, \delta g \right\rangle v_g. \end{aligned}$$

At a point where $\{e_i\}$ is a geodesic orthonormal frame, one has

$$\begin{aligned} 2\operatorname{sym}(\nabla h(d\varphi, \tau(\varphi)))(e_i, e_j) &= 2h(\nabla d\varphi(e_i, e_j), \tau(\varphi)) \\ &\quad + h(d\varphi(e_i), \nabla_{e_j}^\varphi \tau(\varphi)) + h(d\varphi(e_j), \nabla_{e_i}^\varphi \tau(\varphi)), \end{aligned}$$

and

$$\operatorname{div}_M(h(d\varphi, \tau(\varphi))^\sharp) = |\tau(\varphi)|^2 + \langle d\varphi, \nabla^\varphi \tau(\varphi) \rangle.$$

Substitution gives the tensor $S_2(\varphi)$ in (3.29), and hence (3.30) follows. $\square$

Theorem 3.4.5. *Let $\varphi : (M^m, g) \rightarrow (N^n, h)$ be a smooth map. Then*

$$\operatorname{div}_M S_2(\varphi) = -h(\tau_2(\varphi), d\varphi). \quad (3.33)$$

Equivalently,

$$(\operatorname{div}_M S_2(\varphi))(X) = -h(\tau_2(\varphi), d\varphi(X)).$$

Proof. Let $\{e_1, \dots, e_m\}$ be a local orthonormal frame such that $\nabla_{e_i}^M e_j = 0$ at a fixed point $x \in M$. For $X = e_j$, differentiating the expression of $S_2(\varphi)$ gives

$$\begin{aligned} (\operatorname{div}_M S_2(\varphi))(e_j) &= \sum_{i=1}^m e_i(S_2(\varphi)(e_i, e_j)) \\ &= -h(\nabla_{e_j}^\varphi \tau(\varphi), \tau(\varphi)) - e_j \langle d\varphi, \nabla^\varphi \tau(\varphi) \rangle \\ &\quad + \sum_{i=1}^m h(\nabla_{e_i}^\varphi d\varphi(e_i), \nabla_{e_j}^\varphi \tau(\varphi)) + \sum_{i=1}^m h(d\varphi(e_i), \nabla_{e_i}^\varphi \nabla_{e_j}^\varphi \tau(\varphi)). \end{aligned}$$

At the point x ,

$$\tau(\varphi) = \sum_{i=1}^m \nabla_{e_i}^\varphi d\varphi(e_i), \quad \nabla_{e_i}^\varphi d\varphi(e_j) = \nabla_{e_j}^\varphi d\varphi(e_i),$$

and

$$\nabla_{e_i}^\varphi \nabla_{e_j}^\varphi \tau(\varphi) - \nabla_{e_j}^\varphi \nabla_{e_i}^\varphi \tau(\varphi) = R^N(d\varphi(e_i), d\varphi(e_j))\tau(\varphi).$$

Substituting these identities and simplifying yields

$$\operatorname{div}_M S_2(\varphi) = -h(\tau_2(\varphi), d\varphi),$$

which proves the assertion. □

Corollary 3.4.6. *Let $\varphi : (M^m, g) \rightarrow (N^n, h)$ be a smooth map between Riemannian manifolds. Then:*

- (i) *if φ is biharmonic, then $\operatorname{div}_M S_2(\varphi) = 0$;*
- (ii) *if φ is a submersion and $\operatorname{div}_M S_2(\varphi) = 0$, then φ is biharmonic.*

Chapter 4

p -Harmonic and p -Biharmonic Mappings

4.1 The p -energy and p -harmonic maps

In this chapter we recall the variational formalism for p -harmonic maps and then pass to the corresponding p -biharmonic theory. Throughout the chapter,

$$\varphi : (M, g) \longrightarrow (N, h)$$

is a smooth map between Riemannian manifolds, $D \subset M$ is a compact domain, and $p \geq 2$.

Definition 4.1.1. The p -energy of φ on D is the functional $E_p : C^\infty(M, N) \longrightarrow \mathbb{R}_+$ given by

$$E_p(\varphi) = \frac{1}{p} \int_D \|d\varphi\|^p v_g. \quad (4.1)$$

Definition 4.1.2. A smooth map $\varphi : (M^m, g) \rightarrow (N^n, h)$ is called p -harmonic if it is a critical point of the p -energy on every compact domain $D \subset M$. Equivalently, for every compactly supported variation $(\varphi_t)_{t \in (-\varepsilon, \varepsilon)}$ of φ ,

$$\left. \frac{d}{dt} E_p(\varphi_t) \right|_{t=0} = 0. \quad (4.2)$$

4.1.1 First variation of the p -energy

Theorem 4.1.3 (First variation of the p -energy). *Let $\varphi : (M^m, g) \rightarrow (N^n, h)$ be a smooth map and let $(\varphi_t)_{t \in (-\varepsilon, \varepsilon)}$ be a smooth variation of φ supported in D . If*

$$v = \left. \frac{\partial \varphi_t}{\partial t} \right|_{t=0}$$

is the associated variation field, then

$$\left. \frac{d}{dt} E_p(\varphi_t) \right|_{t=0} = - \int_D h(\tau_p(\varphi), v) v_g, \quad (4.3)$$

where the p -tension field is

$$\tau_p(\varphi) = \text{trace}_g \nabla (\|d\varphi\|^{p-2} d\varphi) = \sum_{i=1}^m \left\{ \nabla_{e_i}^\varphi (\|d\varphi\|^{p-2} d\varphi(e_i)) - \|d\varphi\|^{p-2} d\varphi(\nabla_{e_i}^M e_i) \right\}, \quad (4.4)$$

and $\{e_i\}_{i=1}^m$ is a local orthonormal frame on (M, g) .

Proof. Let $\{e_i\}_{i=1}^m$ be an orthonormal frame on (M, g) , and write ∂_t for the standard vector field on $(-\varepsilon, \varepsilon)$. On $M \times (-\varepsilon, \varepsilon)$, equipped with the product metric, the fields $(e_i, 0)$ and $(0, \partial_t)$ satisfy

$$[(e_i, 0), (0, \partial_t)] = 0.$$

Define

$$\Phi : M \times (-\varepsilon, \varepsilon) \longrightarrow N, \quad \Phi(x, t) = \varphi_t(x).$$

Then, at $t = 0$,

$$d\Phi(e_i, 0) = d\varphi(e_i), \quad d\Phi(0, \partial_t) = v.$$

Since

$$\|d\varphi_t\|^p = (\|d\varphi_t\|^2)^{p/2},$$

we get

$$\begin{aligned} \partial_t \|d\varphi_t\|^p &= \frac{p}{2} (\|d\varphi_t\|^2)^{\frac{p}{2}-1} \partial_t \|d\varphi_t\|^2 \\ &= p \|d\varphi_t\|^{p-2} \sum_{i=1}^m h(\nabla_{\partial_t}^\Phi d\Phi(e_i, 0), d\Phi(e_i, 0)) \\ &= p \|d\varphi_t\|^{p-2} \sum_{i=1}^m h(\nabla_{(e_i, 0)}^\Phi d\Phi(\partial_t), d\Phi(e_i, 0)). \end{aligned}$$

At $t = 0$ this becomes

$$\begin{aligned} \partial_t \|d\varphi_t\|^p|_{t=0} &= p \sum_{i=1}^m h(\nabla_{e_i}^\varphi v, \|d\varphi\|^{p-2} d\varphi(e_i)) \\ &= p \sum_{i=1}^m \left\{ e_i(h(v, \|d\varphi\|^{p-2} d\varphi(e_i))) - h(v, \nabla_{e_i}^\varphi (\|d\varphi\|^{p-2} d\varphi(e_i))) \right\}. \end{aligned}$$

Consider the compactly supported one-form

$$\omega(X) = h(v, \|d\varphi\|^{p-2} d\varphi(X)), \quad X \in \Gamma(TM).$$

Its divergence is

$$\begin{aligned} \operatorname{div}^M \omega &= \sum_{i=1}^m (\nabla_{e_i} \omega)(e_i) \\ &= \sum_{i=1}^m \left\{ e_i(h(v, \|d\varphi\|^{p-2} d\varphi(e_i))) - h(v, \nabla_{e_i}^M (\|d\varphi\|^{p-2} d\varphi(e_i))) \right\}. \end{aligned}$$

Therefore

$$\partial_t \|d\varphi_t\|^p|_{t=0} = p \operatorname{div}^M \omega - p \sum_{i=1}^m h(v, \nabla_{e_i}^\varphi (\|d\varphi\|^{p-2} d\varphi(e_i)) - \|d\varphi\|^{p-2} d\varphi(\nabla_{e_i}^M e_i)).$$

After integration over D , the divergence term vanishes by the divergence theorem, since the variation is compactly supported in D . This gives the asserted formula. $\square$

Remark 4.1.4. Assume that $\|d\varphi\|_x \neq 0$ for all $x \in M$. Then φ is p -harmonic if and only if

$$\|d\varphi\|^{p-2} \tau(\varphi) + (p-2) \|d\varphi\|^{p-3} d\varphi(\operatorname{grad}^M \|d\varphi\|) = 0. \quad (4.5)$$

Indeed, with respect to a local orthonormal frame $\{e_i\}_{i=1}^m$,

$$\begin{aligned} \tau_p(\varphi) &= \sum_{i=1}^m \left\{ \nabla_{e_i}^\varphi (\|d\varphi\|^{p-2} d\varphi(e_i)) - \|d\varphi\|^{p-2} d\varphi(\nabla_{e_i}^M e_i) \right\} \\ &= \|d\varphi\|^{p-2} \tau(\varphi) + (p-2) \|d\varphi\|^{p-3} d\varphi(\operatorname{grad}^M \|d\varphi\|). \end{aligned}$$

4.1.2 Second variation of the p -energy

Theorem 4.1.5 (Second variation of the p -energy). *Let $\varphi : (M^m, g) \rightarrow (N^n, h)$ be a p -harmonic map, and let $D \subset M$ be compact. If $(\varphi_{t,s})$ is a two-parameter variation of φ with compact support in D , then*

$$\left. \frac{\partial^2}{\partial t \partial s} E_p(\varphi_{t,s}) \right|_{(t,s)=(0,0)} = \int_D h(J_p^\varphi(v), w) v_g, \quad (4.6)$$

where

$$v = \left. \frac{\partial \varphi_{t,s}}{\partial t} \right|_{t=s=0}, \quad w = \left. \frac{\partial \varphi_{t,s}}{\partial s} \right|_{t=s=0},$$

and the Jacobi operator associated with E_p is

$$J_p^\varphi(v) = -\|d\varphi\|^{p-2} \text{trace}_g R^N(v, d\varphi) d\varphi - \text{trace}_g \nabla^\varphi (\|d\varphi\|^{p-2} \nabla^\varphi v) - (p-2) \text{trace}_g \nabla (h(\nabla^\varphi v, d\varphi) \|d\varphi\|^{p-4} d\varphi). \quad (4.7)$$

Remark 4.1.6. For a local orthonormal frame $\{e_i\}_{i=1}^m$, the three trace terms in (4.7) mean

$$\begin{aligned} \text{trace}_g R^N(v, d\varphi) d\varphi &= \sum_{i=1}^m R^N(v, d\varphi(e_i)) d\varphi(e_i), \\ \text{trace}_g \nabla^\varphi (\|d\varphi\|^{p-2} \nabla^\varphi v) &= \sum_{i=1}^m \left\{ \nabla_{e_i}^\varphi (\|d\varphi\|^{p-2} \nabla_{e_i}^\varphi v) - \|d\varphi\|^{p-2} \nabla_{\nabla_{e_i}^M v}^\varphi v \right\}, \\ \text{trace}_g \nabla (h(\nabla^\varphi v, d\varphi) \|d\varphi\|^{p-4} d\varphi) &= \sum_{i=1}^m \left\{ \nabla_{e_i}^\varphi (h(\nabla^\varphi v, d\varphi) \|d\varphi\|^{p-4} d\varphi(e_i)) - h(\nabla^\varphi v, d\varphi) \|d\varphi\|^{p-4} d\varphi(\nabla_{e_i}^M v) \right\}. \end{aligned}$$

Proof. Let

$$\Phi : M \times (-\varepsilon, \varepsilon) \times (-\varepsilon, \varepsilon) \longrightarrow N, \quad \Phi(x, t, s) = \varphi_{t,s}(x).$$

Let ∇^Φ be the pull-back connection on $\Phi^{-1}TN$. For every vector field X on M ,

$$[\partial_t, X] = 0, \quad [\partial_s, X] = 0, \quad [\partial_t, \partial_s] = 0. \quad (4.8)$$

From the definition of E_p ,

$$\left. \frac{\partial^2}{\partial t \partial s} E_p(\varphi_{t,s}) \right|_{t=s=0} = \frac{1}{p} \int_D \left. \frac{\partial^2}{\partial t \partial s} (\|d\varphi_{t,s}\|^2)^{p/2} \right|_{t=s=0} v_g. \quad (4.9)$$

Choose a geodesic frame $\{e_i\}_{i=1}^m$ at the point under consideration. Since $d\varphi_{t,s}(e_i) = d\Phi(e_i)$, a direct differentiation gives

$$\begin{aligned} \frac{1}{p} \left. \frac{\partial^2}{\partial t \partial s} (\|d\varphi_{t,s}\|^2)^{p/2} \right|_{(t,s)=(0,0)} &= h(\nabla_{\partial_t}^\Phi \nabla_{\partial_s}^\Phi d\Phi(e_i), d\varphi(e_i)) \|d\varphi\|^{p-2} \\ &\quad + h(\nabla_{\partial_s}^\Phi d\Phi(e_i), \nabla_{\partial_t}^\Phi d\Phi(e_i)) \|d\varphi\|^{p-2} \\ &\quad + (p-2) h(\nabla_{\partial_s}^\Phi d\Phi(e_i), d\varphi(e_i)) h(\nabla_{\partial_t}^\Phi d\Phi(e_j), d\varphi(e_j)) \|d\varphi\|^{p-4}. \end{aligned} \quad (4.10)$$

The first term in (4.10) can be rewritten as

$$\begin{aligned} &h(\nabla_{\partial_t}^\Phi \nabla_{\partial_s}^\Phi d\Phi(e_i), d\varphi(e_i)) \|d\varphi\|^{p-2} \\ &= h(R^N(v, d\varphi(e_i)) w, d\varphi(e_i)) \|d\varphi\|^{p-2} + \text{div}^M \omega - h\left(\nabla_{\partial_t}^\Phi d\Phi(\partial_s) \Big|_{(0,0)}, \tau_p(\varphi)\right), \end{aligned}$$

where

$$\omega(X) = h\left(\nabla_{\partial_t}^\Phi d\Phi(\partial_s) \Big|_{(0,0)}, \|d\varphi\|^{p-2} d\varphi(X)\right).$$

The second term satisfies

$$h(\nabla_{\partial_s}^\Phi d\Phi(e_i), \nabla_{\partial_t}^\Phi d\Phi(e_i)) \|d\varphi\|^{p-2} = \operatorname{div}^M \eta - h(w, \nabla_{e_i}^\varphi (\|d\varphi\|^{p-2} \nabla_{e_i}^\varphi v)),$$

with

$$\eta(X) = h(w, \|d\varphi\|^{p-2} \nabla_X^\varphi v).$$

Finally, the last term can be written in the form

$$\begin{aligned} & (p-2)h(\nabla_{\partial_s}^\Phi d\Phi(e_i), d\varphi(e_i))h(\nabla_{\partial_t}^\Phi d\Phi(e_j), d\varphi(e_j)) \|d\varphi\|^{p-4} \\ &= (p-2) \operatorname{div}^M \theta - (p-2)h\left(w, \nabla_{e_i}^\varphi (h(\nabla_{e_j}^\varphi v, d\varphi(e_j)) \|d\varphi\|^{p-4} d\varphi(e_i))\right), \end{aligned}$$

where

$$\theta(X) = h(w, h(\nabla_{e_j}^\varphi v, d\varphi(e_j)) \|d\varphi\|^{p-4} d\varphi(X)).$$

Substitution into (4.10), followed by integration over D , eliminates all divergence terms. Since φ is p -harmonic, $\tau_p(\varphi) = 0$, and the remaining terms are exactly

$$\int_D h(J_p^\varphi(v), w) v_g.$$

This proves the formula. $\square$

4.2 p -biharmonic maps

Definition 4.2.1. The p -bienergy of a smooth map $\varphi : (M^m, g) \rightarrow (N^n, h)$ on a compact domain $D \subset M$ is

$$E_{2,p}(\varphi) = \frac{1}{2} \int_D \|\tau_p(\varphi)\|^2 v_g. \quad (4.11)$$

Definition 4.2.2. A smooth map $\varphi : (M^m, g) \rightarrow (N^n, h)$ is called p -biharmonic if it is a critical point of $E_{2,p}$ on every compact domain $D \subset M$, that is,

$$\left. \frac{d}{dt} E_{2,p}(\varphi_t) \right|_{t=0} = 0 \quad (4.12)$$

for every compactly supported variation (φ_t) of φ .

4.2.1 First variation of the p -bienergy

Theorem 4.2.3 (First variation of the p -bienergy). *Let $\varphi : (M^m, g) \rightarrow (N^n, h)$ be a smooth map, let $(\varphi_t)_{t \in (-\varepsilon, \varepsilon)}$ be a smooth variation supported in D , and assume $p \geq 2$. Then*

$$\left. \frac{d}{dt} E_{2,p}(\varphi_t) \right|_{t=0} = - \int_D h(\tau_{2,p}(\varphi), v) v_g, \quad (4.13)$$

where $v = d\varphi_t/dt|_{t=0}$ and

$$\begin{aligned} \tau_{2,p}(\varphi) &= -\|d\varphi\|^{p-2} \operatorname{trace}_g R^N(\tau_p(\varphi), d\varphi) d\varphi \\ &\quad - \operatorname{trace}_g \nabla^\varphi (\|d\varphi\|^{p-2} \nabla^\varphi \tau_p(\varphi)) \\ &\quad - (p-2) \operatorname{trace}_g \nabla (h(\nabla^\varphi \tau_p(\varphi), d\varphi) \|d\varphi\|^{p-4} d\varphi). \end{aligned} \quad (4.14)$$

Proof. Set $\Phi(x, t) = \varphi_t(x)$. Then

$$\left. \frac{d}{dt} E_{2,p}(\varphi_t) \right|_{t=0} = \int_D h(\nabla_{\partial_t}^{\Phi} \tau_p(\varphi_t), \tau_p(\varphi_t)) v_g \Big|_{t=0}. \quad (4.15)$$

Choose a geodesic frame $\{e_i\}_{i=1}^m$ at the point under consideration. At that point,

$$\nabla_{\partial_t}^{\Phi} \tau_p(\varphi_t) = \nabla_{\partial_t}^{\Phi} \nabla_{e_i}^{\Phi} (\|d\varphi_t\|^{p-2} d\varphi_t(e_i)). \quad (4.16)$$

Using the curvature identity of (N, h) , we obtain

$$\begin{aligned} \nabla_{\partial_t}^{\Phi} \nabla_{e_i}^{\Phi} (\|d\varphi_t\|^{p-2} d\varphi_t(e_i)) &= \|d\varphi_t\|^{p-2} R^N(d\Phi(\partial_t), d\varphi_t(e_i)) d\varphi_t(e_i) \\ &\quad + \nabla_{e_i}^{\Phi} \nabla_{\partial_t}^{\Phi} (\|d\varphi_t\|^{p-2} d\varphi_t(e_i)). \end{aligned} \quad (4.17)$$

Moreover,

$$\begin{aligned} \left. \nabla_{\partial_t}^{\Phi} (\|d\varphi_t\|^{p-2} d\varphi_t(e_i)) \right|_{t=0} &= \|d\varphi\|^{p-2} \nabla_{e_i}^{\varphi} v \\ &\quad + (p-2) h(\nabla_{e_j}^{\varphi} v, d\varphi(e_j)) \|d\varphi\|^{p-4} d\varphi(e_i). \end{aligned} \quad (4.18)$$

An integration by parts, using the compact support of the variation, transforms the second term into the two trace terms appearing in (4.14). Substitution into (4.15) gives the stated formula. $\square$

Theorem 4.2.4. *A smooth map $\varphi : (M^m, g) \rightarrow (N^n, h)$ is p -biharmonic if and only if*

$$\tau_{2,p}(\varphi) = 0, \quad (4.19)$$

that is,

$$\begin{aligned} 0 &= -\|d\varphi\|^{p-2} \text{trace}_g R^N(\tau_p(\varphi), d\varphi) d\varphi - \text{trace}_g \nabla^{\varphi} (\|d\varphi\|^{p-2} \nabla^{\varphi} \tau_p(\varphi)) \\ &\quad - (p-2) \text{trace}_g \nabla (h(\nabla^{\varphi} \tau_p(\varphi), d\varphi) \|d\varphi\|^{p-4} d\varphi). \end{aligned} \quad (4.20)$$

Remark 4.2.5. The p -bitension field can be expressed in terms of the p -Jacobi operator as

$$\tau_{2,p}(\varphi) = J_p^{\varphi}(\tau_p(\varphi)). \quad (4.21)$$

4.3 The stress p -energy tensor

Proposition 4.3.1. *Let $\varphi : (M^m, g) \rightarrow (N^n, h)$ be a smooth map such that $\|d\varphi\|_x \neq 0$ for all $x \in M$, and let $(g_t)_{t \in (-\varepsilon, \varepsilon)}$ be a one-parameter variation of g . Then*

$$\left. \frac{d}{dt} E_p(\varphi) \right|_{t=0} = \frac{1}{2} \int_D \langle S_p(\varphi), \delta g \rangle v_g, \quad (4.22)$$

where $\delta g = \partial_t g_t|_{t=0}$ and the stress p -energy tensor is

$$S_p(\varphi)(X, Y) = \frac{1}{p} \|d\varphi\|^p g(X, Y) - \|d\varphi\|^{p-2} h(d\varphi(X), d\varphi(Y)). \quad (4.23)$$

Theorem 4.3.2. *For every smooth map $\varphi : (M^m, g) \rightarrow (N^n, h)$,*

$$\text{div}^M S_p(\varphi) = -h(\tau_p(\varphi), d\varphi). \quad (4.24)$$

Proof. Let $\{e_i\}_{i=1}^m$ be a local orthonormal frame such that $\nabla_{e_i}^M e_j = 0$ at $x \in M$. Taking X among the frame fields, we compute at x

$$\begin{aligned} (\operatorname{div}^M S_p(\varphi))(X) &= \frac{1}{p} g(X, \operatorname{grad}^M \|d\varphi\|^p) - h(d\varphi(X), d\varphi(\operatorname{grad}^M \|d\varphi\|^{p-2})) \\ &\quad - \|d\varphi\|^{p-2} h(\nabla_X^\varphi d\varphi(e_i), d\varphi(e_i)) - \|d\varphi\|^{p-2} h(d\varphi(X), \tau(\varphi)). \end{aligned}$$

Since

$$\frac{1}{p} X(\|d\varphi\|^p) = \frac{1}{2} \|d\varphi\|^{p-2} X(\|d\varphi\|^2),$$

the gradient terms cancel, and we obtain

$$\begin{aligned} (\operatorname{div}^M S_p(\varphi))(X) &= -h(d\varphi(X), d\varphi(\operatorname{grad}^M \|d\varphi\|^{p-2})) + \|d\varphi\|^{p-2} \tau(\varphi) \\ &= -h(\tau_p(\varphi), d\varphi(X)). \end{aligned}$$

This is precisely (4.24). □

Corollary 4.3.3. *Let $\varphi : (M^m, g) \rightarrow (N^n, h)$ be smooth. Then:*

- (i) *if φ is p -harmonic, then $\operatorname{div}^M S_p(\varphi) = 0$;*
- (ii) *if φ is a submersion and $\operatorname{div}^M S_p(\varphi) = 0$, then φ is p -harmonic.*

4.4 Stress p -bienergy tensors

Let $\varphi : (M, g) \rightarrow (N, h)$ be a smooth map and assume $p \geq 2$. Consider a smooth one-parameter variation of the metric g , namely a family $(g_t)_{t \in (-\varepsilon, \varepsilon)}$ with $g_0 = g$. We write

$$\delta = \left. \frac{\partial}{\partial t} \right|_{t=0}, \quad \delta g \in \Gamma(\odot^2 T^*M).$$

In local coordinates (x^i) on M , if

$$g_t = g_{ij}(t, x) dx^i \otimes dx^j,$$

then

$$\delta g = \left. \frac{\partial g_{ij}}{\partial t} \right|_{t=0} dx^i \otimes dx^j.$$

We start from

$$\left. \frac{d}{dt} E_{2,p}(\varphi) \right|_{t=0} = \frac{1}{2} \int_D \delta(\|\tau_p(\varphi)\|^2) v_g + \frac{1}{2} \int_D \|\tau_p(\varphi)\|^2 \delta(v_{g_t}). \quad (4.25)$$

The computation of the first integral is organized through the following lemmas.

Lemma 4.4.1. *Let*

$$\xi = (\operatorname{div}^M \delta g)^\sharp - \frac{1}{2} \operatorname{grad}^M (\operatorname{trace}_g \delta g). \quad (4.26)$$

Then

$$\begin{aligned} \delta(\|\tau_p(\varphi)\|^2) &= -(p-2) \|d\varphi\|^{p-4} \langle \varphi^* h, \delta g \rangle h(\tau(\varphi), \tau_p(\varphi)) \\ &\quad - 2 \|d\varphi\|^{p-2} \langle h(\nabla d\varphi, \tau_p(\varphi)), \delta g \rangle - 2 \|d\varphi\|^{p-2} h(d\varphi(\xi), \tau_p(\varphi)) \\ &\quad - (p-2)(p-4) \|d\varphi\|^{p-5} \langle \varphi^* h, \delta g \rangle h(d\varphi(\operatorname{grad}^M \|d\varphi\|), \tau_p(\varphi)) \\ &\quad - 2(p-2) \|d\varphi\|^{p-3} \langle d\|d\varphi\| \odot h(d\varphi, \tau_p(\varphi)), \delta g \rangle \\ &\quad - (p-2) \|d\varphi\|^{p-4} h(d\varphi(\operatorname{grad}^M \langle \varphi^* h, \delta g \rangle), \tau_p(\varphi)). \end{aligned} \quad (4.27)$$

Proof. In local coordinates (x^i) on M and (y^α) on N ,

$$\delta(\|\tau_p(\varphi)\|^2) = \delta(\tau_p(\varphi)^\alpha \tau_p(\varphi)^\beta h_{\alpha\beta}) = 2 \delta(\tau_p(\varphi)^\alpha) \tau_p(\varphi)^\beta h_{\alpha\beta}. \quad (4.28)$$

Using the decomposition

$$\tau_p(\varphi)^\alpha = \|d\varphi\|^{p-2} \tau(\varphi)^\alpha + \theta^\alpha, \quad \theta^\alpha = (p-2) \|d\varphi\|^{p-3} g^{ij} (\|d\varphi\|)_i \varphi_j^\alpha, \quad (4.29)$$

we obtain

$$\delta(\tau_p(\varphi)^\alpha) = \delta(\|d\varphi\|^{p-2}) \tau(\varphi)^\alpha + \|d\varphi\|^{p-2} \delta(\tau(\varphi)^\alpha) + \delta(\theta^\alpha). \quad (4.30)$$

The individual terms are

$$\delta(\|d\varphi\|^{p-2}) \tau(\varphi)^\alpha = -\frac{p-2}{2} \|d\varphi\|^{p-4} \langle \varphi^* h, \delta g \rangle \tau(\varphi)^\alpha, \quad (4.31)$$

$$\|d\varphi\|^{p-2} \delta(\tau(\varphi)^\alpha) = -\|d\varphi\|^{p-2} g^{ai} g^{bj} \delta(g_{ab}) (\nabla d\varphi)_{ij}^\alpha - \|d\varphi\|^{p-2} \xi^k \varphi_k^\alpha, \quad (4.32)$$

and

$$\begin{aligned} \delta(\theta^\alpha) = & -\frac{(p-2)(p-3)}{2} \|d\varphi\|^{p-5} \langle \varphi^* h, \delta g \rangle g^{ij} (\|d\varphi\|)_i \varphi_j^\alpha \\ & + (p-2) \|d\varphi\|^{p-3} \delta(g^{ij}) (\|d\varphi\|)_i \varphi_j^\alpha \\ & - \frac{p-2}{2} \|d\varphi\|^{p-4} g^{ij} (\langle \varphi^* h, \delta g \rangle)_i \varphi_j^\alpha + \frac{p-2}{2} \|d\varphi\|^{p-5} g^{ij} (\|d\varphi\|)_i \langle \varphi^* h, \delta g \rangle \varphi_j^\alpha. \end{aligned}$$

Combining these identities and contracting with $2\tau_p(\varphi)^\beta h_{\alpha\beta}$ yields (4.27). $\square$

Lemma 4.4.2. *Let $D \subset M$ be compact. Then*

$$\begin{aligned} \int_D \|d\varphi\|^{p-2} h(d\varphi(\xi), \tau_p(\varphi)) v_g = & \int_D \left\langle -\text{sym}(\nabla(\|d\varphi\|^{p-2} h(d\varphi, \tau_p(\varphi)))) \right. \\ & \left. + \frac{1}{2} \text{div}^M(\|d\varphi\|^{p-2} h(d\varphi, \tau_p(\varphi))^\sharp) g, \delta g \right\rangle v_g. \end{aligned} \quad (4.33)$$

Lemma 4.4.3. *Let*

$$\omega = \|d\varphi\|^{p-4} h(d\varphi, \tau_p(\varphi)).$$

Then

$$-\int_D \|d\varphi\|^{p-4} h(d\varphi(\text{grad}^M \langle \varphi^* h, \delta g \rangle), \tau_p(\varphi)) v_g = \int_D \langle \varphi^* h, \delta g \rangle \text{div}^M \omega v_g. \quad (4.34)$$

Proof. The identity follows from

$$\text{div}^M(\langle \varphi^* h, \delta g \rangle \omega) = \langle \varphi^* h, \delta g \rangle \text{div}^M \omega + \omega(\text{grad}^M \langle \varphi^* h, \delta g \rangle)$$

and the divergence theorem. $\square$

Theorem 4.4.4 (Stress p -bienergy tensor). *Let $\varphi : (M^m, g) \rightarrow (N^n, h)$ be smooth and assume that $\|d\varphi\|_x \neq 0$ for all $x \in M$. If (g_t) is a one-parameter variation of g , then*

$$\left. \frac{d}{dt} E_{2,p}(\varphi) \right|_{t=0} = \frac{1}{2} \int_D \langle S_{2,p}(\varphi), \delta g \rangle v_g, \quad (4.35)$$

where $S_{2,p}(\varphi) \in \Gamma(\odot^2 T^*M)$ is defined by

$$\begin{aligned} S_{2,p}(\varphi)(X, Y) = & -\frac{1}{2} \|\tau_p(\varphi)\|^2 g(X, Y) - \|d\varphi\|^{p-2} h(d\varphi, \nabla^\varphi \tau_p(\varphi)) g(X, Y) \\ & + \|d\varphi\|^{p-2} h(d\varphi(X), \nabla_Y^\varphi \tau_p(\varphi)) + \|d\varphi\|^{p-2} h(d\varphi(Y), \nabla_X^\varphi \tau_p(\varphi)) \\ & + (p-2) \|d\varphi\|^{p-4} h(d\varphi, \nabla^\varphi \tau_p(\varphi)) h(d\varphi(X), d\varphi(Y)). \end{aligned} \quad (4.36)$$

It is called the stress p -bienergy tensor of φ .

Proof. Since

$$\delta(v_{gt}) = \frac{1}{2} \langle g, \delta g \rangle v_g,$$

formula (4.25), together with Lemmas 4.4.1–4.4.3, leads first to the intermediate tensor expression

$$\begin{aligned} S_{2,p}(\varphi) = & -(p-2) \|d\varphi\|^{p-4} h(\tau(\varphi), \tau_p(\varphi)) \varphi^* h - 2 \|d\varphi\|^{p-2} h(\nabla d\varphi, \tau_p(\varphi)) \\ & + 2 \operatorname{sym}(\nabla(\|d\varphi\|^{p-2} h(d\varphi, \tau_p(\varphi)))) \\ & - \operatorname{div}^M(\|d\varphi\|^{p-2} h(d\varphi, \tau_p(\varphi))^\sharp) g \\ & - (p-2)(p-4) \|d\varphi\|^{p-5} h(d\varphi(\operatorname{grad}^M \|d\varphi\|), \tau_p(\varphi)) \varphi^* h \\ & - 2(p-2) \|d\varphi\|^{p-3} d\|d\varphi\| \odot h(d\varphi, \tau_p(\varphi)) \\ & + (p-2) \operatorname{div}^M(\|d\varphi\|^{p-4} h(d\varphi, \tau_p(\varphi))) \varphi^* h + \frac{1}{2} \|\tau_p(\varphi)\|^2 g. \end{aligned} \quad (4.37)$$

For vector fields $X, Y \in \Gamma(TM)$,

$$\begin{aligned} & 2 \operatorname{sym}(\nabla(\|d\varphi\|^{p-2} h(d\varphi, \tau_p(\varphi))))(X, Y) \\ & = 2 \|d\varphi\|^{p-2} h(\nabla d\varphi(X, Y), \tau_p(\varphi)) + \|d\varphi\|^{p-2} h(d\varphi(X), \nabla_Y^\varphi \tau_p(\varphi)) \\ & \quad + \|d\varphi\|^{p-2} h(d\varphi(Y), \nabla_X^\varphi \tau_p(\varphi)) + X(\|d\varphi\|^{p-2}) h(d\varphi(Y), \tau_p(\varphi)) \\ & \quad + Y(\|d\varphi\|^{p-2}) h(d\varphi(X), \tau_p(\varphi)), \end{aligned} \quad (4.38)$$

and

$$\begin{aligned} -2d\|d\varphi\| \odot h(d\varphi, \tau_p(\varphi))(X, Y) = & -X(\|d\varphi\|) h(d\varphi(Y), \tau_p(\varphi)) \\ & - Y(\|d\varphi\|) h(d\varphi(X), \tau_p(\varphi)). \end{aligned} \quad (4.39)$$

In a normal frame at x ,

$$\operatorname{div}^M(\|d\varphi\|^{p-2} h(d\varphi, \tau_p(\varphi))^\sharp) = \|\tau_p(\varphi)\|^2 + \|d\varphi\|^{p-2} h(d\varphi, \nabla^\varphi \tau_p(\varphi)), \quad (4.40)$$

$$\begin{aligned} \operatorname{div}^M(\|d\varphi\|^{p-4} h(d\varphi, \tau_p(\varphi))) = & (p-4) \|d\varphi\|^{p-5} h(d\varphi(\operatorname{grad}^M \|d\varphi\|), \tau_p(\varphi)) \\ & + \|d\varphi\|^{p-4} h(\tau(\varphi), \tau_p(\varphi)) \\ & + \|d\varphi\|^{p-4} h(d\varphi, \nabla^\varphi \tau_p(\varphi)). \end{aligned} \quad (4.41)$$

Substituting these identities into (4.37) gives the simpler expression (4.36). $\square$

Theorem 4.4.5. *Let $\varphi : (M, g) \rightarrow (N, h)$ be smooth and assume that $\|d\varphi\|_x \neq 0$ for all $x \in M$. Then*

$$(\operatorname{div}^M S_{2,p}(\varphi))(X) = -h(\tau_{2,p}(\varphi), d\varphi(X)), \quad X \in \Gamma(TM). \quad (4.42)$$

Proof. Recall that

$$\begin{aligned} \tau_{2,p}(\varphi) = & -\|d\varphi\|^{p-2} \operatorname{trace}_g R^N(\tau_p(\varphi), d\varphi) d\varphi - \operatorname{trace}_g \nabla^\varphi(\|d\varphi\|^{p-2} \nabla^\varphi \tau_p(\varphi)) \\ & - (p-2) \operatorname{trace}_g \nabla(h(\nabla^\varphi \tau_p(\varphi), d\varphi) \|d\varphi\|^{p-4} d\varphi), \end{aligned}$$

and

$$\tau_p(\varphi) = \operatorname{div}^M(\|d\varphi\|^{p-2} d\varphi). \quad (4.43)$$

Choose a local orthonormal frame $\{e_i\}_{i=1}^m$ such that $\nabla_{e_i}^M e_j = 0$ at x . A direct computation of $\sum_i e_i(S_{2,p}(\varphi)(e_i, X))$, using the curvature identity

$$\nabla_{e_i}^\varphi \nabla_X^\varphi \tau_p(\varphi) - \nabla_X^\varphi \nabla_{e_i}^\varphi \tau_p(\varphi) = R^N(d\varphi(e_i), d\varphi(X)) \tau_p(\varphi), \quad (4.44)$$

shows that all remaining terms collect into

$$-h(\tau_{2,p}(\varphi), d\varphi(X)).$$

This proves (4.42). $\square$

Remark 4.4.6. When $p = 2$, the tensor $S_{2,p}(\varphi)$ reduces to the classical stress bienergy tensor $S_2(\varphi)$.

Corollary 4.4.7. *Let $\varphi : (M, g) \rightarrow (N, h)$ be smooth.*

- (1) *If $S_{2,m}(\varphi) = 0$, where $m = \dim M$, then φ is m -harmonic.*
- (2) *If M is compact without boundary and $p \neq m/2$, then $S_{2,p}(\varphi) = 0$ implies that φ is p -harmonic.*

Proof. Let $\{e_i\}_{i=1}^m$ be a local orthonormal frame. Taking the trace of $S_{2,p}$ gives

$$0 = \sum_{i=1}^m S_{2,p}(\varphi)(e_i, e_i) = -\frac{m}{2} \|\tau_p(\varphi)\|^2 + (p - m) \|d\varphi\|^{p-2} h(d\varphi, \nabla^\varphi \tau_p(\varphi)). \quad (4.45)$$

For $p = m$, this becomes

$$-\frac{m}{2} \|\tau_m(\varphi)\|^2 = 0,$$

and hence φ is m -harmonic.

For the second assertion, set

$$\theta(X) = h(\|d\varphi\|^{p-2} d\varphi(X), \tau_p(\varphi)), \quad X \in \Gamma(TM).$$

The trace identity can be written as

$$0 = \left(\frac{m}{2} - p\right) \|\tau_p(\varphi)\|^2 + (p - m) \operatorname{div}^M \theta. \quad (4.46)$$

Integrating over the compact boundaryless manifold M and using Green's theorem yields

$$\left(\frac{m}{2} - p\right) \int_M \|\tau_p(\varphi)\|^2 v_g = 0. \quad (4.47)$$

Since $p \neq m/2$, we conclude that $\tau_p(\varphi) = 0$; thus φ is p -harmonic. $\square$

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